KHALIFA UNIVERSITY

COSC 336 Introduction to Software Engineering

Phase 3
Requirements Specification

Intelligent AI Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

KU ResourceSavers | Group 7 | 14 October 2026

This Phase 3 report defines 158 functional requirements, 31 nonfunctional requirements and 41 user goals. It includes two completed demonstration stakeholder interviews, system comparisons, requirement traceability and verification criteria.

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Status Requirements specification complete for team review; two demonstration stakeholder interviews completed. Instructor deadline confirmation, team sign-off and later application testing remain pending.

Key workflow rule An allocation request holds no stock on submission. Final source approval creates the reservation after destination acceptance.

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Reading route: sections 1-3 establish meaning and boundaries; sections 4-5 contain the SRS requirements; sections 6-7 model user goals; sections 8-10 provide priorities, evidence and references. Appendix A links each requirement to its source, use cases, prerequisites and acceptance check. Appendix B provides a readable actor index for the overview diagram.

1 Introduction

KU ResourceSavers is an academic prototype for university departments to discover, request and reuse assets while retaining custody, maintenance and sustainability evidence. This requirements document defines what the software must do and how its behavior will be verified. It supports later design, implementation and testing; it does not claim that the application or acceptance tests are complete.

This specification contains 158 functional requirements, 31 nonfunctional requirements and 41 detailed use cases. Sections 4 and 5.1 to 5.4 follow the supplied SRS guidance, and sections 6 and 7 provide one complete use-case overview and detailed descriptions. Completed demonstration interviews and their findings appear in section 9.2; appendices A and B contain the supporting records. [S1, S9, S12-S13]

1.1 Evidence and document conventions

S1 and S9 provide the project brief and Phase 3 submission structure. S2-S5 and S10-S11 identify supporting reports, feedback and teaching materials; source availability is recorded in section 10.1. S6-S8 identify the existing-system comparison sources. S12-S13 record two completed demonstration stakeholder interviews on 12 October 2026. Demonstration answers are not university policy.

Shall states required behavior. FR, NFR, UC, BR, AC and D identify functional requirements, quality constraints, use cases, business rules, acceptance checks and open decisions. High/P0 means build the required foundation first; Medium/P1 means complete required intelligence and reporting after that foundation. Desirable/Low extensions and Future items do not remove required baseline functions.

Each requirement contains one shall obligation and one corresponding acceptance check. Letter-suffixed IDs are separate requirements with their own AC IDs. A data-record schema, eligibility predicate or atomic transaction is verified as one behavior; its fields or invariants are not separate workflow actions.

1.2 Structure and source alignment

- The report follows S9 slide 26 and SRS sections 4 and 5.1 to 5.4.
- One complete overview diagram represents all 41 detailed user goals (Figure 1).
- Use cases identify description, primary actor, supporting actors, preconditions, trigger, numbered main flow, alternatives, exceptions, both outcomes and rules.
- A pending allocation request is non-binding. Final source approval creates the stock hold after destination acceptance.
- The 48-hour window applies to pending allocation requests, not an existing stock hold.
- Every FR and NFR, including letter-suffixed IDs, has one corresponding acceptance check. Appendix A lists all requirement IDs. NFR-16 and NFR-16a specify safety controls.
- The conflicting 5 October and 18 October deadlines are recorded without assuming an extension. No actual campus interviews, human approvals or passed application tests are claimed.

Lecturer instruction	Location in this report
Purpose and introduction	Section 1
SRS 4 System features	Section 4; each feature has description/priority, stimulus/response and FRs
SRS 5.1 Performance	Section 5.1
SRS 5.2 Safety	Section 5.2
SRS 5.3 Security	Section 5.3
SRS 5.4 Software quality	Section 5.4
One overview diagram and detailed use cases	Sections 6 and 7
Priorities and traceability	Section 8

Lecturer instruction	Location in this report
Evidence, unresolved dates and handover	Section 9

NFR-17 specifies fairness checks, FR-14.1f covers assistant navigation, and FR-15.1–FR-15.1g define reuse indicators. Section 9.2 documents the completed demonstration interviews. External confirmations and application test results remain pending where evidence has not been collected.

1.3 Existing system comparison

The comparison reviews official public descriptions, not a hands-on trial. It informs requirements without proving campus demand or claiming that an unmentioned vendor feature is absent.

System and source	Documented capability	Implication for this specification
Snipe-IT [S6]	Asset assignment, location, history including maintenance, requestable items and granular role permissions.	Keep asset identity, custody, life-cycle history and record-level access explicit: FR-01, FR-02, FR-08 and FR-09.
Warp It [S7]	Surplus listing and search, organizational reuse, impact reporting, and advertised AI listing assistance and transfer auditability.	Specify publication, requests, transfers and evidenced impact. AI is not claimed as unique: FR-03 to FR-07, FR-10, FR-15 and FR-16.
Rheaply [S8]	Visibility of existing resources, exchange of surplus, and reporting dashboards for reuse outcomes.	Connect available inventory with purchasing needs and explain reporting evidence: FR-05, FR-11, FR-15 and FR-17.

1.4 Classic and AI enhanced approaches

The prototype combines deterministic controls with advisory AI. The comparison below defines the intended added value and limitations; measured improvement will be established during later evaluation, using the same data and permission scope for both approaches. [S1]

Area	Classic approach	AI approach and oversight
Search and matching	Keyword and structured filters return assets matching explicit terms and recorded fields. Results are predictable but can miss equivalent descriptions.	Semantic matching can connect different wording and rank suitable alternatives. Permissions, safety, availability and mandatory specifications are checked first; users may choose ordinary search or another eligible asset.
Asset classification	The custodian selects a category and enters metadata manually, with field validation.	AI suggests category, tags, description and missing metadata. Incorrect or unsupported suggestions remain possible; the custodian accepts, edits or rejects them before saving.
Sustainable actions	Recorded condition, costs, safety rules and human approval govern service and disposition.	AI compares retention, reuse, repair and recovery options, explains evidence gaps and proposes actions. It cannot certify safety, authorize release or execute disposition.
Reports	Saved records and fixed formulas produce reproducible operational totals and qualified benefit estimates.	AI summarizes the same verified data and may describe trends or propose actions. Numeric claims must match source reports and saved summaries require human review.
Trade-offs	Deterministic rules are preferred for authorization, quantities, state transitions and calculations because these must be consistent and auditable.	AI may improve discovery and readability but introduces uncertainty, bias, privacy and availability risks. Recorded model metadata, review, restricted inputs and manual fallback limit these risks.

Evaluation will compare keyword and semantic retrieval on the same ten held-out requests, report suitable results in the top three, latency and failures, and review explanation usefulness. Reviewers will record recommendation acceptance or override without assuming that acceptance proves correctness. Conventional and AI-written reports will be checked against identical underlying figures. No improvement or accuracy result is claimed in this requirements phase.

2 Scope actors and business rules

2.1 Included scope

- Accounts, departments, scoped roles and authenticated access.
- Asset registration, optional photos/evidence, publication, search and requests.
- Reservations, distinct departmental approvals, dispatch, receipt and dispute records.
- Inspection, maintenance, repair and authorized donation, recycling, retirement or disposal records.
- AI-assisted classification, semantic matching, ranked explanations, sustainable-action suggestions, a limited assistant and report summaries.
- In-application notifications, protected audit history, sustainability indicators and operational reports.

2.2 Excluded scope

- Real payment processing or automated purchasing.
- Live integration with university identity, procurement or asset systems without institutional approval.
- A separate native mobile application.
- Physical transport, repair work, safety certification or waste handling performed by the software. The system records those activities.
- Production deployment across the university or production-scale service guarantees.
- Training a new large language model or proving campus-wide AI accuracy.
- Unapproved use of confidential university records.
- Automatic approval or execution of transfers, financial decisions or disposal by AI.

2.3 Operating context and assumptions

The prototype is an English-language browser application. The proposed Phase 2 stack is Django, HTML/CSS with limited JavaScript and MySQL, with a hosted AI connection. These are feasibility choices subject to a technical trial; exact deployment and model versions belong in the later design/configuration records. The requirement to use Git and the shared GitHub repository applies throughout development. [S1, S3, S4]

Sample data will contain 200 assets in five categories and four fictional departments, 40 requests, 20 transfer examples and 20 maintenance examples. The test fixture may include separate role accounts for the same scenario, but no student should approve their own request through a second assigned role on the same account. All real-campus authority and data access remain subject to confirmation.

Default categories are Furniture, Computers, Electronics, Laboratory Equipment and Books or Office Materials. Category-specific specifications and mandatory units will be frozen with the evaluation dataset. Data categories may be refined after review of the demonstration findings or later campus validation without weakening required coverage.

2.4 Demand allocation and reservation meaning

This prototype distinguishes a general resource need from a request for particular stock. A demand record states what a department needs and supports AI matching or partial fulfilment from several assets. An allocation request selects an asset and quantity for that demand. Its submission and destination acceptance hold no units. Final source-custodian approval creates the reservation and moves Free quantity to Held atomically.

Record or stage	State and stock effect
Resource demand	Open or PartiallyFulfilled until receipt-confirmed quantity meets the need; no stock effect by itself.
Allocation request submission	PendingApproval; validate current Free quantity but reserve nothing.
Destination acceptance	PendingApproval with acceptance recorded; reserve nothing.
Final source approval	Allocation Approved and one linked reservation Approved; Free becomes Held once.
Transfer creation and dispatch	Creation converts the existing reservation with no new deduction; dispatch moves Held to InTransit.
Receipt	Full acceptance completes transfer and moves destination stock to InUse; discrepancy remains Disputed.

S11 uses the word request for the asset-specific operation represented here by allocation request. The eight-monitor example therefore behaves identically at the stock boundary: submission leaves eight Free; successful final custodian approval of five leaves three Free and five Held. The additional destination acceptance stage and general demand record are project refinements. The 48-hour Phase 2 proposal is retained as a pending-response window, not a pre-approval hold. Campus authority and the chosen window still require stakeholder confirmation. [S3, S11]

2.5 Actors and permissions

Role	Permitted records	Actions	Restriction
Requester	Own requests/reservations; permitted campus listings and asset history	Create/amend/cancel demand; submit/cancel allocation requests; cancel own pre-dispatch reservations; review AI	Cannot approve own request or inspect restricted finance/audit details
Department representative	Assigned department listings, requests and workflow status	Coordinate department requests; listing changes only with separately assigned custodian permission	No implicit release, receipt or finance approval
Asset custodian	Assigned assets, relevant requests and transfers	Register/publish/withdraw; authorize source release; arrange/dispatch transfer; propose and execute approved disposition	Cannot approve own request or act as both decision-makers; no direct workflow-balance edits
Receiving custodian	Assigned destination transfer and relevant asset details	Confirm receipt or report discrepancy	Cannot rewrite source approval or bypass a dispute
Destination approver	Assigned destination-department acceptance queue	Accept/reject destination need before source-custodian review	Cannot be the requester or the final source decision-maker
Maintenance staff	Assigned task, asset condition and service history	Inspect; record work/cost; record fit-for-use decision	No unrelated asset, ownership or disposal approval
Procurement officer	Permitted requests, alternatives and purchase-need evidence	Record purchasing-need evidence and assess reuse options	Finance review is still required for counted avoided purchases
Finance officer	Assigned financial evidence and permitted report totals	Review replacement cost, transfer/repair cost and qualified savings	Cannot change completed transfers through financial review
Sustainability officer	Permitted life-cycle and environmental records	Review mass, factors, baselines; propose actions; inspect indicators	No automatic authority to execute disposal

Role	Permitted records	Actions	Restriction
Asset-management administrator	Authorized campus workflow and audit scope	Resolve disputes; approve consequential disposition; inspect audit records	Must be distinct from the proposer; cannot silently erase history
System administrator	Accounts, role configuration and technical operation	Create/deactivate accounts; configure access; operate backup/quota settings	No business approval authority merely from technical-admin status

A requester or representative acts within destination scope. Campus user is the general actor role used by sign-in, search, defect reporting and history; those actions enforce their individual permissions. Report viewer means an account granted report scope, which may also hold finance or sustainability roles. External AI is a supporting actor with no business-approval authority. Each actor appears once in Figure 1; Appendix B provides the consolidated actor index.

2.6 Business rules

BR-01 Least privilege Access depends on role, department and assigned record. System-administrator status alone grants no release, financial or disposal approval. All UI, download and AI paths share the same checks.

BR-02 Preserved history Identifiers remain stable. Referenced records are deactivated or archived rather than hard-deleted through ordinary workflows. Corrections append an event with reason; they do not erase prior decisions.

BR-03 Conserved quantities Each tracked unit occupies exactly one bucket: Free, Held, InTransit, InUse, Blocked or Retired. Registration creates InUse or Blocked units; publication moves eligible InUse units to Free. Pending allocation requests consume no stock. Final approval alone moves Free to Held; transfer creation reuses that hold. Split lots retain parent links and disposition preserves historical quantities.

BR-04 Safety and eligibility Only published Active stock in Good or Fair condition and the Free bucket is reservable. NeedsInspection, NeedsRepair and Unsafe conditions prohibit allocation. A new safety issue moves affected unheld stock to Blocked; existing Held/InTransit quantities retain their bucket but cannot dispatch or complete normally until cleared or resolved.

BR-05 Pending expiry and hold release A PendingApproval allocation request expires 48 elapsed hours after submission and has no held stock to release. Final approval creates an Approved reservation with no automatic expiry. Pre-dispatch cancellation releases Held once: to Free if published and eligible, to Blocked if clearance is missing or unsafe, otherwise to InUse. Dispatch ends this cancellation route.

BR-06 Demand accounting Remaining demand equals requested minus receipt-confirmed quantity. Committed quantity equals approved holds plus unresolved in-transit/disputed quantities, without counting converted reservations twice. Pending requests are non-binding. Submission and final approval check uncommitted demand; only approval commits it. Core demand fields are locked while pending requests or commitments exist. Fulfilment or cancellation closes remaining pending requests without stock movement.

BR-07 Transfer completion Both departmental approvals precede dispatch. Legal university ownership remains unchanged; departmental responsibility and custody change at full accepted receipt. Dispatch moves Held to InTransit; receipt creates destination InUse holdings. Repeated events have no second effect.

BR-08 Approval order and separation The assigned destination department approver records acceptance first. The assigned source asset custodian records release second and triggers atomic reservation. These two decision-makers must be distinct and neither may be the requester. Source approval rechecks all eligibility and quantities; failed validation leaves the allocation request pending with no new hold.

BR-09 Disposition and disputes A distinct asset-management administrator authorizes consequential disposition. A custodian records its execution. Receipt discrepancies remain Disputed until documented accepted receipt or return; disputed stock earns no reuse benefit.

BR-10 AI advises AI proposes; users confirm. Mandatory specifications and safety override semantic similarity. Scores are not calibrated probabilities. Explicit user confirmation does not bypass permissions, stock rules or human approvals.

BR-11 Manual continuity AI failure, timeout, quota exhaustion or insufficient confidence does not disable ordinary forms, keyword search, human approvals or conventional reports. No AI output alone changes authoritative asset state.

BR-12 Evidence and uncertainty Unknown data remains unknown. Figures and policy answers identify their evidence and boundaries. External AI receives only permitted minimized inputs. Model output is not evidence of equipment safety or university policy.

BR-13 Qualified benefits Financial benefit requires completed reuse and distinct validated purchasing-need evidence. Waste diversion requires mass and an otherwise-discarded baseline. One need allocation or disposal episode is counted once; corrections remain traceable.

BR-14 Reliable notifications Notifications follow recorded business events. Delivery failures do not undo valid decisions. Event-recipient identity prevents duplicate notifications on retries; users only see permitted detail.

BR-15 Prototype timestamps Store timestamps in UTC, display an explicit timezone and use Asia/Dubai for user-facing dates. Pending allocation requests expire after 48 elapsed hours, not rounded calendar days. Approved reservations have no automatic time expiry.

2.7 Allowed state transitions

Each change rechecks current state, authority and quantity at save time. Successful important changes include their audit event; notification delivery can retry separately. Unlisted ordinary transitions are denied. Terminal history is retained and corrections require an authorized, reasoned event.

Demand transitions

Current state	Event	Result
New	Valid demand submission	Open; received zero; no stock movement
Open or PartiallyFulfilled	Accepted receipt below total	PartiallyFulfilled; add received quantity once
Open or PartiallyFulfilled	Accepted receipts reach total	Fulfilled; close leftover pending allocation requests
Open or PartiallyFulfilled	Cancel with no transit/dispute	Cancelled remainder; cancel pending requests and release approved pre-dispatch holds once

Allocation request transitions

Current state	Event	Result
None	Submit within current stock/demand	PendingApproval; no stock movement
PendingApproval	Destination accepts	Same state; acceptance saved; no stock movement
PendingApproval with acceptance	Final source approval after recheck	Approved; create one Approved reservation and move Free to Held
PendingApproval	Rejection, requester cancellation or 48-hour expiry	Rejected, Cancelled or Expired; no stock release
PendingApproval	Insufficient stock at final approval	Stay pending with error; retry before expiry or cancel

Reservation transitions

Current state	Event	Result
None	Successful final source approval	Approved; one quantity hold

Current state	Event	Result
Approved	Create transfer	Converted; same hold now controlled by linked transfer
Approved or Converted before dispatch	Authorized cancellation	Cancelled; release once according to BR-05
Approved or Converted	Original request expiry time passes	No automatic expiry of the approved hold

Transfer transitions

Current state	Event	Result
None	Create from Approved reservation	Approved; no second deduction
Approved	Valid dispatch	InTransit; Held to InTransit; source custody retained
Approved	Pre-dispatch cancellation	Cancelled; linked hold released once
InTransit	Full acceptable receipt	Completed; destination InUse; demand received updated
InTransit	Quantity or condition discrepancy	Disputed; no benefit awarded
Disputed	Documented full accepted receipt	Completed; destination InUse or Blocked according to condition
Disputed	Documented complete return	Returned; source InUse or Blocked; demand commitment released

Publication and maintenance transitions

Current state	Event	Result
Draft or Withdrawn	Publish eligible InUse units	Published; selected quantity becomes Free
Published	Withdraw Free units	Quantity becomes InUse; Withdrawn when none remains offered
Eligible unheld stock	Safety defect	Blocked; existing holds/transit retained but restricted
No task	Defect report	Reported
Reported	Start inspection	UnderInspection
UnderInspection	Start service	InProgress
UnderInspection or InProgress	Final condition and clearance decision	Completed; eligible unheld Blocked to InUse, otherwise Blocked
Reported or UnderInspection	Cancel duplicate task with reason	Cancelled; no automatic safety clearance

Disposition transitions

Current state	Event	Result
None	Human proposal	Proposed; no stock movement
Proposed	Distinct administrator decision	Approved or Rejected; no execution

Current state	Event	Result
Proposed or Approved	Cancel before execution	Cancelled; no stock movement
Approved	Record authorized actual execution	Completed; affected units Retired with specific life-cycle outcome

2.8 Notification recipients

Event	Recipients	Required effect
Allocation submitted	Requester; destination approver	Confirmation and pending decision
Destination accepted	Requester; source custodian	Source review task; no hold yet
Final source approval	Requester; source and receiving custodians	Approved reservation reference
Rejected cancelled or expired	Requester; assigned approvers/custodians	Reason/status; release only an existing approved hold
Dispatch	Requester; receiving custodian	InTransit status
Receipt or dispute	Requester; source and receiving custodians; asset administrator for dispute	Completion evidence or unresolved discrepancy
Maintenance update	Reporter; custodian; assigned staff	Task and safety-state update

3 Data interfaces and indicator definitions

3.1 Data dictionary

Record or fields	Required content or allowed values	Validation and meaning
Asset identity	asset_id generated; name 1-120 characters; description up to 2000; category from controlled list; optional subcategory/tags	Unique immutable ID. External asset tag optional but unique when entered.
Asset specifications	Named value/unit pairs; mandatory request specifications flagged separately	Canonical units per category; unknown mandatory specifications cannot qualify as a match.
Asset responsibility	Owning department, assigned custodian, location; all required	Departments/users must be active for new assignments. Legal owner remains the university.
Asset quantity	Tracking mode Individual or Lot; positive integer; Individual always one	Lots must be homogeneous in condition/specification/custody; split holdings retain parent IDs.
Asset condition	Good, Fair, NeedsInspection, NeedsRepair, Unsafe	Only Good/Fair are ordinarily allocatable; condition is independent from stock bucket.
Asset life cycle	Active, Retired, Donated, Recycled, Disposed	External disposition is terminal in the ordinary workflow; correction requires an auditable authorized resolution.
Asset publication and stock	Draft/Published/Withdrawn; Free/Held/InTransit/InUse/Blocked/Retired quantities	Each unit occupies one bucket. A zero-Free listing is not reservable even when still Published.

Record or fields	Required content or allowed values	Validation and meaning
Dates and values	Acquisition date, estimated value in AED, mass in kg, optional supporting evidence	Dates no later than today for acquisition; non-negative entered values; Unknown distinct from zero.
Attachments	JPEG/PNG photos or PDF evidence; at most five per asset; 5 MiB each	Validate file contents and size; protected storage/download; reject other types.
Resource demand	ID, requester, department, category, purpose, specs, positive quantity, minimum condition, destination, urgency, required date	Urgency Low/Normal/High; minimum condition Good or Fair; required date today or later in Asia/Dubai. This record alone does not select or hold stock.
Allocation and reservation	Allocation ID, demand/asset IDs, quantity, requester, pending expiry, destination/source decisions; linked reservation ID only after final approval	Pending expiry is 48 elapsed hours; no hold before final approval; distinct decision-makers cannot be requester.
Transfer	Reservation ID, source/destination, quantity, arrangements, approvals, dispatch/receipt times, receiver, condition, evidence	One active transfer per reservation; full receipt or documented dispute; no silent partial completion.
Maintenance	Task ID, asset, reporter, assignee, findings, actions, parts, costs, service dates, final condition, fit-for-use decision	Costs non-negative; completion cannot precede start; final clearance required to restore eligibility.
AI record	Task and request ID, permitted input references, output, provider/model, available version, configuration, time, outcome, reviewer/override	Do not log credentials or unnecessary personal data; unavailable provider version explicitly labelled.
Benefit evidence	Completed event, need allocation ID, baseline and price evidence, costs, quantity, mass, carbon factor/source/year/original and converted units/geography/boundary, reviewer	Separate simulated/estimated/confirmed values; count qualifying need allocations or disposal episodes once.
Audit and notification	Event ID, target, actor, UTC time, action/outcome; notification recipient and read/delivery state	Append-only audit through application roles; unique event-recipient delivery identity.

3.2 User and service interfaces

Users require role-appropriate views for asset entry/publication, search/results, requests, reservations, approval queues, handover/receipt, maintenance, recommendations, reports, notifications and account administration. Error messages identify the rejected field or business rule and preserve valid unsaved input where safe. Wireframe styling and detailed API contracts are deferred to Phases 4 and 5.

The AI interface accepts only permitted minimized asset/request data or verified aggregate report figures and returns suggestions plus task metadata. Missing, malformed, timed-out and over-quota responses are explicit outcomes. Unsupported file content is reported instead of assumed analyzed. The application validates every proposed write after human confirmation.

There is no required specialist hardware interface. A browser and the selected hosting environment form the prototype operating boundary. Sample database imports, AI configuration, protected attachment storage, notification processing and backup/restore procedures will be documented during design and implementation. Integration choices must preserve this specification even if a proposed vendor changes.

3.3 AI factor and evidence contract

Stage	Requirement boundary
Eligibility	Apply permissions, Active/Published state, usable condition, quantity and mandatory specification checks before scoring. Unknown mandatory values are excluded.
Ranking	Record purpose/category/specification fit, quantity, condition, location, urgency/date, cost and available sustainability factors. Missing optional factors are disclosed rather than invented.
Scoring	Expose a documented compatibility scale and configuration version. The detailed formula/weights are a Phase 5 design choice; stored factor values and configuration must explain the saved order.
Tie handling	Use a documented stable tie rule, such as asset ID, after equivalent factors. Do not introduce unexplained departmental preference.
Human review	Show reasons, uncertainty and alternatives. Record acceptance or override. Confidence is unavailable unless its meaning is documented.
Evaluation	Compare keyword and semantic retrieval on the same held-out requests; record improvement and failures without asserting campus-wide accuracy.

3.4 Sustainability indicators

An activity count is not automatically a benefit. Each indicator has a reporting period, unit, evidence requirement and calculation version. Reports distinguish fictional fixture results, estimated real-world values and reviewed observed records. Unknown inputs remain unknown and the dashboard reports the number of eligible records with complete evidence. [S1, S3]

Indicator	Definition and counting rule	Required evidence
Completed reuse units	Sum of quantities in Completed internal transfers in the period. This is movement activity; a later completed movement of the same item is another event, not a new avoided purchase by default.	Completed transfer IDs and receipt dates; show distinct asset IDs separately.
Reuse rate for a publication cohort	100 x units from a defined first-publication cohort received for reuse by the cutoff / total units first offered in that cohort. Count each unit once; zero denominator is Not applicable.	Cohort start/end, cutoff date, offered quantities and traceable unit/lot movements.
Repair completions	Count Completed maintenance tasks with a recorded repair action and final condition during the period. Report inspected-only tasks separately.	Task ID, work record and completion date.
Avoided purchase value	Qualified received quantity x finance-reviewed replacement unit price. Count only the allocation to a distinct confirmed purchasing need.	Need/evidence ID, quantity, replacement quote or recorded basis, finance reviewer and date.
Net reuse benefit	Avoided purchase value minus attributed transfer costs minus attributed repair costs. Keep negative results visible. Software project cost is reported separately.	Qualified avoided purchase plus complete attributed cost records in AED.
Waste diversion in kg	Sum of qualified unit mass x qualifying quantity for a documented otherwise-discarded episode. A later movement does not recount that same episode.	Mass source, disposal baseline, unique episode reference and completed reuse/repair evidence.

Indicator	Definition and counting rule	Required evidence
Estimated life extension	Revised expected end-of-use date minus the documented baseline expected end-of-use date, expressed in months using the stated day-to-month convention. Label Estimated; show observed continued use separately.	Baseline/revised dates, reviewer, estimation basis; use days/30.4375 for months.
Estimated avoided emissions in kg CO2e	For equivalent functional service and a stated boundary: baseline scenario emissions minus reuse/repair scenario emissions. Include added transport/repair impacts; retain negative values.	Cited factor/year, original and converted units, mass, displacement fraction, added impacts, boundary and reviewer. Desktop demonstration factor: section 3.4.3 [S16]; unsupported categories remain Unknown.

3.4.1 Worked illustrative example

Assume 20 chairs satisfy a confirmed purchasing need at AED 150 each. Avoided purchase value is 20 x AED 150 = AED 3000. With AED 400 transport and AED 300 repair, net reuse benefit is AED 2300 before software costs. If each chair is 8 kg and the documented alternative was disposal, the same episode represents 160 kg of waste diversion. These figures reproduce the Phase 2 planning example; they are hypothetical, not measured KU outcomes. Carbon remains Unknown without an appropriate sourced factor and boundary.

3.4.2 Data completeness and correction

If replacement cost is known but repair cost is unknown, show known gross avoided purchase value and Unknown net benefit. Do not replace unknown cost with zero. If an event is corrected, retain the previous calculation and reviewer history, update its current qualification and recompute totals without creating a second benefit. Cross-report filters must use the same event dates and qualification rules.

3.4.3 Carbon factor and calculation boundary

Selected demonstration factor: U.S. EPA Waste Reduction Model (WARM), December 2023, Electronics, Exhibit 1-5: desktop CPUs, source reduction, -20.86 metric tonnes CO2e per short ton. Desktop CPUs exclude monitors and peripherals. The negative source value denotes an emissions reduction; this report uses its positive magnitude as an avoided-production proxy. [S16]

Unit conversion: 20.86 × 1000 / 907.18474 = 22.994214 kg CO2e per kg of desktop CPU mass. This U.S.-based factor is not a UAE-specific factor. Use it only for reviewed desktop reuse that displaces equivalent new production; document the displacement fraction and functional-service assumption. Do not apply it to chairs, monitors or unrelated categories. [S16]

Illustrative fixture: 10 kg of desktop CPU mass, displacement fraction 1.0, and 10 kg CO2e of separately documented additional transport/repair impacts gives (10 × 22.994214 × 1.0) − 10 = 219.94 kg CO2e estimated avoided emissions. The added-impact value is a hypothetical test input, not a published factor or measured campus result. Missing displacement, mass, added-impact evidence or category fit produces Unknown. The result is a scenario estimate, not a full product life-cycle inventory or certified offset.

Each factor record includes publisher, document/table, factor year, original value/unit, converted unit, material class, geography, calculation boundary and reviewer. The sustainability reviewer confirms applicability before an estimate is counted.

4 System features

This section follows SRS section 4. Each feature states its description and priority, stimulus/response sequences and uniquely identified functional requirements. The main and exception flows in section 7 expand the interaction sequences. High/P0 and Medium/P1 both remain essential to the final academic prototype; lower-priority extensions are listed separately in section 8. Every acceptance check is planned, not executed.

4.1 Accounts departments and permissions

4.1.1 Description and priority

Establish authenticated accounts, departments and explicit authority without losing historical identities.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 3-4 and 8; S2 FR-01; S3 sections 3 and 9. Detailed use cases: UC-01, UC-02, UC-03, UC-04, UC-05.

4.1.2 Stimulus and response sequences

Actor action or stimulus	System response
User supplies credentials	Validate active identity and create a scoped session; deny invalid credentials.
Administrator confirms an account, role or department change	Validate authority/references and save an audited change.

4.1.3 Functional requirements

FR-01.1 The system shall allow a system administrator to create prototype accounts.

AC-FR-01.1 Create a valid account as administrator and verify that the stored account can be retrieved. Rules: BR-01.

FR-01.1a The system shall allow a system administrator to deactivate an account.

AC-FR-01.1a Deactivate an active account and verify that its next authenticated action is denied. Rules: BR-01.

FR-01.1b The system shall allow a system administrator to assign an account to its department and explicit roles.

AC-FR-01.1b Save a role/department assignment and verify that the resulting access matches the permission matrix. Rules: BR-01.

FR-01.1c The system shall disable public self-registration.

AC-FR-01.1c Attempt unauthenticated account registration and verify that no account is created. Rules: BR-01.

FR-01.2 The system shall authenticate active users with valid credentials.

AC-FR-01.2 Submit valid active-user credentials and verify a session; invalid or inactive-user credentials produce no session.

FR-01.2a The system shall invalidate the authenticated session on logout.

AC-FR-01.2a Log out and reuse the old session for a protected action; access is denied.

FR-01.3 The system shall enforce the role and record-scope permissions in section 2 on every protected read, download and change, including direct requests and assistant actions.

AC-FR-01.3 Exercise each permitted action and a denied counterpart using two departments; unauthorized attempts reveal no protected record and make no change. Rules: BR-01.

FR-01.4 The system shall preserve historical references to deactivated accounts and departments.

AC-FR-01.4 Deactivate a referenced former user and verify that their historical audit identity remains retrievable. Rules: BR-02.

FR-01.4a The system shall prevent department deactivation while holdings or workflows remain unresolved.

AC-FR-01.4a Attempt to deactivate a department with an active transfer and verify a reasoned refusal. Rules: BR-02.

4.2 Asset records and validation

4.2.1 Description and priority

Maintain complete asset records, validated quantities and protected supporting evidence.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 3 and 8; S2 FR-02; S3 sections 3.4 and 4. Detailed use cases: UC-06, UC-07, UC-22, UC-41.

4.2.2 Stimulus and response sequences

Actor action or stimulus	System response
Custodian supplies or edits asset fields	Validate the data dictionary and attachments.
Custodian confirms valid data	Save identity/history; registration alone does not publish stock.

4.2.3 Functional requirements

FR-02.1 The system shall assign each asset an immutable unique identifier.

AC-FR-02.1 Register two assets and attempt to overwrite their identifiers; the distinct original identifiers remain unchanged.

FR-02.1a The system shall store asset records according to the data dictionary, including its required-field and unknown-value rules.

AC-FR-02.1a Submit a dictionary fixture and verify exact field persistence; omission of a required field rejects that fixture.

FR-02.2 The system shall allow a custodian to update descriptive information for an assigned asset.

AC-FR-02.2 Edit an assigned asset description and retrieve the saved text. Rules: BR-02, BR-03.

FR-02.2a The system shall preserve audited prior values when asset descriptions change.

AC-FR-02.2a Change a description and verify that the previous and current values are identifiable in history. Rules: BR-02, BR-03.

FR-02.2b The system shall reject direct asset-form edits to workflow-controlled quantities or custody.

AC-FR-02.2b Attempt each protected field edit through the asset form and verify that the stored workflow values remain unchanged. Rules: BR-02, BR-03.

FR-02.3 The system shall validate asset quantities as positive integers.

AC-FR-02.3 Submit quantity boundary cases 0, 0.5 and 1; only 1 is accepted. Rules: BR-15.

FR-02.3a The system shall reject negative entered monetary values.

AC-FR-02.3a Submit costs of -1 and 0 AED; only 0 is accepted. Rules: BR-15.

FR-02.3b The system shall reject negative entered mass.

AC-FR-02.3b Submit masses of -1 and 0 kg; only 0 is accepted. Rules: BR-15.

FR-02.3c The system shall reject duplicate external asset tags when a tag is supplied.

AC-FR-02.3c Submit an existing tag on a second asset and verify rejection without creating the record. Rules: BR-15.

FR-02.3d The system shall reject acquisition dates later than the current user-facing date.

AC-FR-02.3d Submit tomorrow's acquisition date in Asia/Dubai and verify rejection. Rules: BR-15.

FR-02.4 The system shall represent individual equipment with quantity one.

AC-FR-02.4 Register individual equipment with quantity two and verify rejection. Rules: BR-03, BR-07.

FR-02.4a The system shall record traceable quantity movements for homogeneous low-value lots.

AC-FR-02.4a Complete a four-unit movement from a ten-unit lot and trace the source and destination quantities to that movement. Rules: BR-03, BR-07.

FR-02.4b The system shall conserve the parent quantity and parent identifier when a lot is split.

AC-FR-02.4b Split ten units into six and four; verify the parent link and a combined quantity of ten. Rules: BR-03, BR-07.

FR-02.5 The system shall accept optional asset attachments only when their verified content is JPEG, PNG or PDF.

AC-FR-02.5 Upload each allowed content type and a renamed executable; only allowed content is accepted. Rules: BR-01.

FR-02.5a The system shall reject an asset attachment larger than 5 MiB.

AC-FR-02.5a Upload files of exactly 5 MiB and 5 MiB plus one byte; only the first is accepted. Rules: BR-01.

FR-02.5b The system shall limit each asset to five attachments.

AC-FR-02.5b Add a sixth attachment and verify rejection while the existing five remain. Rules: BR-01.

FR-02.5c The system shall restrict attachment retrieval to authorized users.

AC-FR-02.5c Request a protected attachment using an unrelated account and verify that no file bytes are returned. Rules: BR-01.

4.3 Publication and availability

4.3.1 Description and priority

Publish usable surplus and withdraw unheld units while preserving commitments.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 3 and 8; S2 FR-03; S3 sections 6 and 9.5. Detailed use cases: UC-08, UC-09.

4.3.2 Stimulus and response sequences

Actor action or stimulus	System response
Custodian selects quantity to publish or withdraw	Check current eligibility and the relevant stock bucket.
Custodian confirms	Move eligible units once and refresh visibility; preserve existing holds.

4.3.3 Functional requirements

FR-03.1 The system shall let a custodian publish an assigned active asset only when its required fields are complete, its offered quantity is available and its condition is Good or Fair.

AC-FR-03.1 Publish a complete Good-condition lot; reject missing fields, excess offered quantity, NeedsInspection and NeedsRepair. Rules: BR-03, BR-04.

FR-03.2 The system shall display current reservable quantity and publication state to authorized campus users.

AC-FR-03.2 View a partly held listing and verify its displayed Free quantity and publication state against the stored record. Rules: BR-04.

FR-03.2a The system shall exclude withdrawn or ineligible assets from ordinary search and AI candidates.

AC-FR-03.2a Query a fixture containing withdrawn and unsafe assets through both retrieval paths; neither asset is a candidate. Rules: BR-04.

FR-03.3 The system shall permit a custodian to withdraw unheld quantity.

AC-FR-03.3 Withdraw three Free units from a partly held lot and verify that those three units are no longer published. Rules: BR-04, BR-05.

FR-03.3a The system shall preserve existing holds after withdrawal until explicit cancellation or resolution.

AC-FR-03.3a Withdraw the Free portion of a held lot and verify that its reservation and Held quantity remain unchanged. Rules: BR-04, BR-05.

FR-03.3b The system shall suspend dispatch when a new safety issue affects the asset.

AC-FR-03.3b Record a safety issue before dispatch and verify that dispatch is denied. Rules: BR-04, BR-05.

4.4 Departmental resource requests

4.4.1 Description and priority

Record departmental needs and track fulfilment independently of stock commitments.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 3 and 8; S2 FR-04; S3 sections 4.1 and 6. Detailed use cases: UC-11, UC-12, UC-13, UC-14, UC-22.

4.4.2 Stimulus and response sequences

Actor action or stimulus	System response
Requester confirms a new demand	Validate fields and create Open demand without holding stock.
Requester amends, cancels or inspects progress	Apply commitment restrictions and separate received from reserved quantity.

4.4.3 Functional requirements

FR-04.1 The system shall record a request identifier, requester, department, category, purpose, mandatory specifications, quantity, minimum condition, destination, urgency and required date.

AC-FR-04.1 Submit a complete request and retrieve the same fields; reject omitted required values and a required date before today. Rules: BR-15.

FR-04.2 The system shall permit an authorized requester or representative to amend Open or PartiallyFulfilled demand only when no pending allocation, approved hold, in-transit or disputed transfer is linked.

AC-FR-04.2 Attempt a specification amendment for each linked-state fixture; only uncommitted demand accepts the amendment. Rules: BR-06.

FR-04.2a The system shall identify the linked records that block a demand amendment.

AC-FR-04.2a Attempt amendment with a pending allocation and verify that the refusal identifies that allocation. Rules: BR-06.

FR-04.3 The system shall derive demand state from receipt-confirmed quantity under BR-06.

AC-FR-04.3 Confirm receipt of four units against demand for ten and verify PartiallyFulfilled with four received. Rules: BR-06, BR-07.

FR-04.3a The system shall report reserved and in-transit quantities separately from fulfilled quantity.

AC-FR-04.3a Reserve and dispatch four units without receipt and verify that fulfilled quantity remains zero. Rules: BR-06, BR-07.

FR-04.4 The system shall reject cancellation of demand linked to an InTransit or Disputed transfer.

AC-FR-04.4 Attempt cancellation in each prohibited state and verify that the demand remains active. Rules: BR-05, BR-06.

FR-04.4a The system shall cancel pending allocation requests without stock movement when their demand is cancelled.

AC-FR-04.4a Cancel eligible demand with a pending allocation and verify that the allocation closes with unchanged stock. Rules: BR-05, BR-06.

FR-04.4b The system shall release approved pre-dispatch holds exactly once under BR-05 when their demand is cancelled.

AC-FR-04.4b Cancel demand containing a hold and retry cancellation; verify one release to the BR-05 bucket. Rules: BR-05, BR-06.

FR-04.4c The system shall preserve prior receipt quantities when outstanding demand is cancelled.

AC-FR-04.4c Cancel partly fulfilled eligible demand and verify that its receipt-confirmed quantity is unchanged. Rules: BR-05, BR-06.

4.5 Search and filtering

4.5.1 Description and priority

Find eligible published assets through permitted keyword and category searches.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 3 and 8; S2 FR-05; S3 sections 2.4 and 3.6. Detailed use cases: UC-10.

4.5.2 Stimulus and response sequences

Actor action or stimulus	System response
User submits category, keyword and filters	Return only permitted eligible results with current Free quantity.
No result or search failure occurs	Offer revised criteria or retry; change no stock.

4.5.3 Functional requirements

FR-05.1 The system shall provide case-insensitive keyword search within the user's permission scope.

AC-FR-05.1 Search a known title in upper and lower case and verify identical permitted result IDs. Rules: BR-01.

FR-05.1a The system shall apply combined category, condition, department, location, available-quantity, asset-age, estimated-value and life-cycle-status filters within the user's permission scope.

AC-FR-05.1a Run a fixed filter fixture covering every filter and their conjunction; every returned row satisfies all selected predicates. Rules: BR-01.

FR-05.2 The system shall show asset identifier, name, category, condition, reservable quantity, department and location in search results.

AC-FR-05.2 Retrieve a known result and compare its displayed fields with the asset record.

FR-05.2a The system shall label unknown search-result values explicitly as Unknown.

AC-FR-05.2a Search an asset with no acquisition date and verify Unknown age.

FR-05.2b The system shall display a no-results response for an empty result set.

AC-FR-05.2b Search for a nonexistent asset and verify no-results guidance.

FR-05.3 The system shall keep ordinary search available when AI is unavailable.

AC-FR-05.3 Disable AI and verify successful keyword retrieval of a known asset. Rules: BR-11.

FR-05.3a The system shall identify the retrieval method for displayed results.

AC-FR-05.3a Display keyword and semantic result fixtures and verify the corresponding method labels. Rules: BR-11.

4.6 Allocation requests and reservations

4.6.1 Description and priority

Submit non-binding allocation requests and reserve quantity only at successful final approval.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 p 8; S2 FR-06; S3 sections 3.4 and 11.4; S9-S11. Detailed use cases: UC-13, UC-14, UC-15, UC-16, UC-18, UC-19, UC-20.

4.6.2 Stimulus and response sequences

Actor action or stimulus	System response
Requester submits a specified asset quantity	Check current stock and demand; record PendingApproval with no hold.

Actor action or stimulus	System response
Final source approval succeeds	Atomically recheck and create one Approved reservation; move Free to Held.
Pending expiry or cancellation occurs	Close the pending request with no stock movement; approved cancellation releases an existing hold once.

4.6.3 Functional requirements

FR-06.1 The system shall create a uniquely identified PendingApproval allocation linking one open demand, one published eligible asset lot and a positive integer quantity within current Free stock and uncommitted demand.

AC-FR-06.1 Submit a valid five-unit allocation against eight Free units and retrieve the complete pending record; invalid quantity fixtures create no record. Rules: BR-03, BR-06.

FR-06.1a The system shall preserve stock, custody and location when an allocation request is submitted.

AC-FR-06.1a Submit five units against eight Free units and compare the pre/post stock, custody and location snapshot; it is unchanged. Rules: BR-03, BR-06.

FR-06.2 The system shall revalidate permission, publication, condition, Free stock and uncommitted demand at final source approval.

AC-FR-06.2 Alter each prerequisite after submission and attempt approval; the invalid fixture remains pending with no hold. Rules: BR-03.

FR-06.2a The system shall atomically commit final approval, one linked Approved reservation and the Free-to-Held movement only after successful revalidation.

AC-FR-06.2a Concurrently approve two five-unit requests against eight Free units; exactly one five-unit hold commits and three units remain Free. Rules: BR-03.

FR-06.3 The system shall store requester, demand and asset IDs, quantity, submission/expiry times, decisions and state for each allocation request.

AC-FR-06.3 Retrieve a pending allocation and compare every required field with its submitted and recorded values. Rules: BR-15.

FR-06.3a The system shall store a distinct linked reservation ID, approval time and held quantity after successful final approval.

AC-FR-06.3a Approve a valid request and verify one reservation whose quantity and time match the committed approval. Rules: BR-15.

FR-06.4 The system shall expire a still-pending allocation at 48 elapsed hours without stock movement.

AC-FR-06.4 Advance a pending fixture to 48 hours and verify Expired with unchanged stock. Rules: BR-05, BR-15.

FR-06.4a The system shall reject approval of a pending request at or beyond its expiry time.

AC-FR-06.4a Attempt approval at the 48-hour boundary and verify refusal without a reservation. Rules: BR-05, BR-15.

FR-06.4b The system shall keep an Approved reservation held without automatic expiry.

AC-FR-06.4b Advance an approved fixture beyond 48 hours and verify that its hold remains. Rules: BR-05, BR-15.

FR-06.5 The system shall permit the requester to cancel a pending allocation without stock movement.

AC-FR-06.5 Cancel a pending five-unit request against eight Free units and verify eight Free units remain. Rules: BR-05.

FR-06.5a The system shall permit the requester or assigned source custodian to cancel a pre-dispatch reservation with exactly one release under BR-05.

AC-FR-06.5a Cancel a held fixture twice and verify a single release to the required bucket. Rules: BR-05.

FR-06.5b The system shall refuse reservation cancellation after dispatch.

AC-FR-06.5b Attempt cancellation of an InTransit transfer's reservation and verify no state or quantity change. Rules: BR-05.

4.7 Approvals transfers and receipt

4.7.1 Description and priority

Record destination acceptance, source release, transfer arrangements, dispatch, receipt and disputes.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 3 and 8; S2 FR-07; S3 sections 6 and 11.5; S9-S11. Detailed use cases: UC-14, UC-17, UC-18, UC-19, UC-20, UC-21, UC-22, UC-23.

4.7.2 Stimulus and response sequences

Actor action or stimulus	System response
Destination approver accepts; source custodian then approves	Record both distinct decisions; final source approval creates the hold.
Custodian arranges and dispatches transfer	Reuse the hold, then move Held to InTransit with source custody retained.
Receiver confirms or reports a discrepancy	Complete full receipt or record Disputed until evidenced resolution.

4.7.3 Functional requirements

FR-07.1 The system shall enforce destination acceptance before source-custodian release approval.

AC-FR-07.1 Attempt source approval before destination acceptance and verify refusal. Rules: BR-08.

FR-07.1a The system shall create the reservation only at successful final source release under FR-06.2 and FR-06.2a.

AC-FR-07.1a Accept at destination then approve at source; verify that only the latter event creates a hold. Rules: BR-08.

FR-07.1b The system shall require a reason for rejection at either pending approval stage.

AC-FR-07.1b Submit an empty rejection reason at each stage and verify rejection of that decision. Rules: BR-08.

FR-07.1c The system shall preserve stock and custody when an allocation is rejected.

AC-FR-07.1c Reject a pending allocation and verify the pre/post stock and custody snapshots are identical. Rules: BR-08.

FR-07.2 The system shall require distinct source and destination decision-makers, neither being the requester.

AC-FR-07.2 Exercise requester self-approval and reuse of the destination approver as source approver; neither decision is accepted. Rules: BR-08, BR-15.

FR-07.2a The system shall record each decision with actor, scope, timestamp and applicable rejection reason.

AC-FR-07.2a Save a valid rejection and verify every decision field against the fixture. Rules: BR-08, BR-15.

FR-07.2b The system shall revalidate the decision-maker's role when a decision is saved.

AC-FR-07.2b Revoke a role after opening the decision form and verify that saving is denied. Rules: BR-08, BR-15.

FR-07.3 The system shall create at most one active transfer per reservation.

AC-FR-07.3 Submit transfer creation twice for one reservation and verify exactly one active transfer. Rules: BR-03, BR-07.

FR-07.3a The system shall preserve the existing hold, approvals and links when creating a transfer.

AC-FR-07.3a Create a transfer and compare its reservation/approval references and unchanged Held quantity with the source record. Rules: BR-03, BR-07.

FR-07.3b The system shall record collection or delivery details and required documents for a transfer.

AC-FR-07.3b Save a complete transfer arrangement fixture and retrieve the same details and document references. Rules: BR-03, BR-07.

FR-07.4 The system shall atomically move the eligible transfer quantity from Held to InTransit on authorized source dispatch.

AC-FR-07.4 Dispatch a valid transfer twice and verify a single Held-to-InTransit movement. Rules: BR-04, BR-07.

FR-07.4a The system shall record the actual dispatch timestamp.

AC-FR-07.4a Dispatch a fixture with a controlled clock and verify its saved UTC timestamp. Rules: BR-04, BR-07.

FR-07.4b The system shall retain source custody until valid receipt.

AC-FR-07.4b Dispatch a transfer without receipt and verify that source custody is unchanged. Rules: BR-04, BR-07.

FR-07.5 The system shall let the assigned receiving custodian confirm the full dispatched quantity and condition, completing the transfer and updating departmental ownership, custody, destination, received quantity and history as one change.

AC-FR-07.5 Confirm a valid receipt and inspect every affected record; repeat the same receipt and verify no duplicate movement or fulfillment. Legal ownership remains with the university. Rules: BR-07.

FR-07.6 The system shall mark a transfer Disputed when receipt quantity or condition differs from dispatch.

AC-FR-07.6 Report a receipt discrepancy and verify Disputed rather than Completed. Rules: BR-07, BR-09.

FR-07.6a The system shall preserve unresolved disputed quantity outside completed reuse.

AC-FR-07.6a Report missing units and verify that the unresolved quantity remains traceable with no reuse benefit. Rules: BR-07, BR-09.

FR-07.6b The system shall require documented asset-management-administrator resolution before disputed completion or confirmed return.

AC-FR-07.6b Attempt closure without authorized resolution, then with a documented resolution; only the latter closes the dispute. Rules: BR-07, BR-09.

4.8 Life cycle history and disposition

4.8.1 Description and priority

Preserve life-cycle history and separate disposition proposal, approval and actual execution.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 3 and 8; S2 FR-08; S3 sections 6 and 9. Detailed use cases: UC-06, UC-23, UC-25, UC-27, UC-28, UC-29, UC-41.

4.8.2 Stimulus and response sequences

Actor action or stimulus	System response
Custodian or sustainability officer proposes an action	Record Proposed with reasons; make no stock movement.
Distinct administrator approves and custodian later records execution	Validate each stage and update life cycle once, retaining history.

4.8.3 Functional requirements

FR-08.1 The system shall retain chronological acquisition, allocation, use, inspection, maintenance, repair, transfer, refurbishment, donation, recycling, retirement and disposal events with actor, time and related record.

AC-FR-08.1 Inspect a fixture asset across these event types and follow links to the originating records; missing optional events are not invented. Rules: BR-02.

FR-08.2 The system shall let an authorized custodian or sustainability officer propose retention, repair, refurbishment, parts recovery, donation, recycling, retirement or disposal with a reason and supporting evidence.

AC-FR-08.2 Submit a disposition proposal linked to the asset and inspect its action, reason and evidence; no asset state changes on proposal alone. Rules: BR-09.

FR-08.3 The system shall require a distinct authorized asset-management administrator to approve donation, parts recovery, recycling, retirement or disposal, with no unresolved holds or transfers and with required safety clearance.

AC-FR-08.3 Deny self-approval and an action against held stock; approve an eligible proposal without yet recording execution. Rules: BR-04, BR-09.

FR-08.4 The system shall record approved disposition execution with quantity, date, responsible custodian and destination or handler evidence.

AC-FR-08.4 Execute a complete approved fixture and verify its stored execution fields. Rules: BR-02, BR-03, BR-09.

FR-08.4a The system shall update quantities and life-cycle status exactly once for a disposition execution.

AC-FR-08.4a Execute a two-unit recycling action twice and verify one two-unit stock adjustment. Rules: BR-02, BR-03, BR-09.

FR-08.4b The system shall preserve prior asset history after disposition execution.

AC-FR-08.4b Execute disposition and verify that all preceding event IDs remain retrievable. Rules: BR-02, BR-03, BR-09.

4.9 Inspection maintenance and repair

4.9.1 Description and priority

Record defects, service work and human clearance before restoring eligibility.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 3-4 and 8; S2 FR-09; S3 sections 6 and 9.5. Detailed use cases: UC-24, UC-25, UC-26, UC-27.

4.9.2 Stimulus and response sequences

Actor action or stimulus	System response
Permitted user reports a defect	Create an inspection task and apply required safety restrictions.
Assigned staff records work and final clearance	Preserve findings/costs; close only with condition and a fit-for-use decision.

4.9.3 Functional requirements

FR-09.1 The system shall allow a permitted user to record an asset defect with description, time and reporter.

AC-FR-09.1 Submit a defect and verify the linked record and its required fields. Rules: BR-04.

FR-09.1a The system shall create an inspection task for a reported defect.

AC-FR-09.1a Record one defect and verify one linked inspection task. Rules: BR-04.

FR-09.1b The system shall block new allocation for an asset with an unresolved safety-related defect.

AC-FR-09.1b Report a safety defect and verify refusal of a new allocation. Rules: BR-04.

FR-09.1c The system shall block dispatch for an asset with an unresolved safety-related defect.

AC-FR-09.1c Report a safety defect on held stock and verify refusal of dispatch. Rules: BR-04.

FR-09.2 The system shall let assigned maintenance staff record inspection findings, work status, actions, parts, costs and service dates.

AC-FR-09.2 Save a complete assigned-task service entry and verify every entered field. Rules: BR-01, BR-02.

FR-09.2a The system shall preserve earlier maintenance entries when new entries are recorded.

AC-FR-09.2a Add a second service entry and verify that the first entry remains unchanged. Rules: BR-01, BR-02.

FR-09.3 The system shall require final condition and an explicit fit-for-use decision before maintenance closure.

AC-FR-09.3 Attempt closure with either field missing and verify rejection. Rules: BR-04.

FR-09.3a The system shall permit republication after maintenance only with authorized clearance and ordinary eligibility.

AC-FR-09.3a Attempt publication without clearance, then with clearance and eligible condition; only the latter succeeds. Rules: BR-04.

FR-09.4 The system shall route an uneconomical or unsuccessful repair to a human-reviewed action proposal without automatically retiring or disposing of the asset.

AC-FR-09.4 Record an unsuccessful repair and generate a linked proposal; verify that stock is blocked and no disposal executes. Rules: BR-09, BR-10.

4.10 AI assisted classification

4.10.1 Description and priority

Suggest asset metadata from supported input with custodian review.

Priority: Medium / P1. Scope: Essential. This completes required intelligence or reporting after trustworthy records and manual workflows exist. Source: S1 pp 4 and 8; S2 FR-10; S3 sections 3.5 and 4. Detailed use cases: UC-30.

4.10.2 Stimulus and response sequences

Actor action or stimulus	System response
Custodian requests suggestions	Send minimized permitted input and display supported outputs with limitations.
Custodian accepts, edits or rejects	Save only user-confirmed valid fields; preserve manual entry on failure.

4.10.3 Functional requirements

FR-10.1 The classification service shall use approved asset title, description, specifications and supported photo or document content to suggest category, subcategory, tags, standardized description, material and missing metadata.

AC-FR-10.1 Run labelled text, image and PDF examples; return the stated fields where supported and mark unavailable outputs explicitly rather than inventing them. Rules: BR-10, BR-12.

FR-10.2 The system shall display classification suggestions beside existing values.

AC-FR-10.2 Open a suggestion fixture and verify that both original and proposed values are visible. Rules: BR-10.

FR-10.2a The system shall save only classification values explicitly accepted or corrected by an authorized custodian.

AC-FR-10.2a Reject one suggested value and accept a corrected value; verify the saved record contains only the authorized change. Rules: BR-10.

FR-10.3 The system shall flag unreadable, unsupported, contradictory or insufficient classification input.

AC-FR-10.3 Submit one fixture for each input defect and verify an explicit reason for each failed classification. Rules: BR-11.

FR-10.3a The system shall permit manual asset completion after classification failure.

AC-FR-10.3a Force classification failure and complete registration using the ordinary form. Rules: BR-11.

4.11 Semantic matching and ranking

4.11.1 Description and priority

Rank eligible assets by semantic suitability and documented factors without allocating them.

Priority: Medium / P1. Scope: Essential. This completes required intelligence or reporting after trustworthy records and manual workflows exist. Source: S1 pp 4 and 8; S2 FR-11; S3 sections 3.5 and 4.5. Detailed use cases: UC-31.

4.11.2 Stimulus and response sequences

Actor action or stimulus	System response
Requester asks for matches to demand	Filter eligibility before semantic ranking and display up to three results.
Requester selects an alternative	Allow an allocation request; do not reserve stock until final approval.

4.11.3 Functional requirements

FR-11.1 The system shall exclude unauthorized, unavailable, unsafe and mandatory-specification-incompatible assets before semantic ranking, including assets with unknown mandatory specifications until reviewed.

AC-FR-11.1 A request requiring 16 GB RAM excludes 8 GB and unknown-RAM items; restricted or held-out stock cannot appear as a valid match. Rules: BR-01, BR-04, BR-10.

FR-11.2 The matching service shall compare the meaning of eligible asset descriptions with the request and return up to three highest-ranked candidates, or fewer when fewer are eligible.

AC-FR-11.2 Use a request phrased differently from asset keywords and inspect semantic candidates; with two eligible items, return two without padding results. Rules: BR-10.

FR-11.3 The system shall rank eligible candidates using declared factors for purpose, category, specifications, quantity, condition, distance or location, urgency, required date, transfer and repair cost, and sustainability evidence.

AC-FR-11.3 Inspect saved factor values and configuration; recompute the saved order from those factors and explain a supported preference change without requiring identical repeated model output. Rules: BR-10, BR-12.

FR-11.4 The system shall let a requester compare recommended candidates with ordinary search results.

AC-FR-11.4 Open both result sets for a demand fixture and verify permitted candidate details are available for comparison. Rules: BR-10.

FR-11.4a The system shall let the requester select an eligible alternative to the top recommendation.

AC-FR-11.4a Select the second eligible candidate and verify that it populates the allocation draft. Rules: BR-10.

FR-11.4b The system shall preserve stock before final approval of an AI-assisted allocation.

AC-FR-11.4b Select and submit an alternative candidate and verify that Free quantity remains unchanged. Rules: BR-10.

4.12 AI explanations oversight and records

4.12.1 Description and priority

Explain AI output, record uncertainty and model metadata, and enforce human authority.

Priority: Medium / P1. Scope: Essential. This completes required intelligence or reporting after trustworthy records and manual workflows exist. Source: S1 pp 4-6 and 8; S2 FR-12; S3 sections 4.5 and 6.5. Detailed use cases: UC-30, UC-31, UC-32, UC-33, UC-38.

4.12.2 Stimulus and response sequences

Actor action or stimulus	System response
An AI task returns an output	Present reasons, evidence, confidence meaning and available version metadata.
User confirms or overrides	Record the outcome and revalidate through normal permissions and business rules.

4.12.3 Functional requirements

FR-12.1 The system shall display recommendation reasons grounded in permitted record fields.

AC-FR-12.1 Trace every specification claim in a recommendation fixture to an accessible asset field. Rules: BR-10.

FR-12.1a The system shall display known limitations beside a recommendation.

AC-FR-12.1a Supply a fixture with missing optional evidence and verify its limitation is displayed. Rules: BR-10.

FR-12.1b The system shall label compatibility scores as ranking scores rather than calibrated probabilities.

AC-FR-12.1b Display a score of 87 and verify that it is not described as 87 percent certainty. Rules: BR-10.

FR-12.2 The system shall display calibrated confidence only when its method and meaning are recorded, otherwise showing calibrated confidence as unavailable.

AC-FR-12.2 Remove the confidence-method record and verify an unavailable label rather than a percentage. Rules: BR-10.

FR-12.2a The system shall identify missing evidence relevant to the confidence assessment.

AC-FR-12.2a Remove a required evidence reference and verify that the missing evidence is named. Rules: BR-10.

FR-12.3 The system shall record AI request identifier, task, permitted input references, provider/model identifier, available version, configuration version, time, outcome and user acceptance or override.

AC-FR-12.3 Retrieve a recommendation record and identify the model and review outcome; where a provider version is unavailable, retain the reported model identifier and mark the version unavailable. Rules: BR-02, BR-12.

FR-12.4 The system shall prevent AI from executing protected approvals, ownership changes, unsafe allocation or disposition.

AC-FR-12.4 Attempt protected actions through AI and direct AI-action endpoints; verify no protected state change. Rules: BR-08, BR-10.

FR-12.4a The system shall revalidate user-confirmed AI-assisted actions through ordinary business rules.

AC-FR-12.4a Confirm an AI draft whose stock became unavailable and verify ordinary validation rejects it. Rules: BR-08, BR-10.

4.13 Sustainable action suggestions

4.13.1 Description and priority

Recommend supported reuse, repair or recovery actions for human review.

Priority: Medium / P1. Scope: Essential. This completes required intelligence or reporting after trustworthy records and manual workflows exist. Source: S1 pp 4 and 8; S2 FR-13; S3 sections 6.5 and 11.4. Detailed use cases: UC-32.

4.13.2 Stimulus and response sequences

Actor action or stimulus	System response
Actor asks for the next sustainable action	Use known condition, demand and cost/evidence inputs; expose uncertainty.
Actor accepts a suggestion	Create a proposal only; actual approval and execution use normal workflows.

4.13.3 Functional requirements

FR-13.1 The service shall suggest retention, transfer, repair, refurbishment, donation, parts recovery, recycling, retirement or disposal using recorded condition, demand, costs, safety restrictions and available sustainability evidence.

AC-FR-13.1 Review usable, repairable and unsafe sample assets; each suggestion identifies the considered alternatives and supporting or missing evidence. Rules: BR-09, BR-10.

FR-13.2 The system shall present suggested action, reasons, uncertainties and alternatives for human review.

AC-FR-13.2 Open a repairable-asset recommendation and verify the complete review fields against the evidence fixture. Rules: BR-09, BR-10.

FR-13.2a The system shall present disposal as a last option when recorded reuse or recovery alternatives are unsuitable.

AC-FR-13.2a Provide a viable repair alternative and verify disposal is not the preferred action. Rules: BR-09, BR-10.

4.14 Natural language assistant

4.14.1 Description and priority

Answer permitted questions and prepare explicitly confirmed demand drafts.

Priority: Medium / P1. Scope: Essential. This completes required intelligence or reporting after trustworthy records and manual workflows exist. Source: S1 pp 4 and 8; S2 FR-14; S3 sections 3.6 and 6.5. Detailed use cases: UC-33.

4.14.2 Stimulus and response sequences

Actor action or stimulus	System response
User asks a question or requests a draft	Retrieve permitted facts or prepare editable fields with references.
User confirms a corrected draft	Validate and submit ordinary demand; no automatic approval or hold.

4.14.3 Functional requirements

FR-14.1 The assistant shall support asset search using permitted application records.

AC-FR-14.1 Ask for a known asset and verify the response identifies an accessible matching record. Rules: BR-01, BR-12.

FR-14.1a The assistant shall prepare resource-request drafts from user input.

AC-FR-14.1a Ask for five chairs and verify a draft for five chairs without a saved request. Rules: BR-01, BR-12.

FR-14.1b The assistant shall answer request-status questions from permitted records.

AC-FR-14.1b Ask about a known request and verify the answer matches its current stored state. Rules: BR-01, BR-12.

FR-14.1c The assistant shall answer asset-history questions from permitted records.

AC-FR-14.1c Ask for an asset's last transfer and verify the cited event matches its history. Rules: BR-01, BR-12.

FR-14.1d The assistant shall answer policy questions from approved reference content.

AC-FR-14.1d Ask a supported policy question and verify the cited approved passage supports the answer. Rules: BR-01, BR-12.

FR-14.1e The assistant shall explain sustainability indicators from their definitions and permitted evidence.

AC-FR-14.1e Ask how net benefit was calculated and verify the response matches the linked source calculation. Rules: BR-01, BR-12.

FR-14.1f The assistant shall provide navigation to authorized system functions.

AC-FR-14.1f Ask how to submit a request, inspect history and view reports; each permitted link opens its function, and forbidden functions are not exposed. Rules: BR-01, BR-12.

FR-14.2 The assistant shall present request drafts with editable fields.

AC-FR-14.2 Correct the draft quantity and verify that the displayed draft retains the correction. Rules: BR-06, BR-10.

FR-14.2a The system shall require explicit confirmation before submitting an assistant-prepared request.

AC-FR-14.2a Prepare a draft without confirming and verify that no request is stored. Rules: BR-06, BR-10.

FR-14.2b The system shall apply standard request validation to confirmed assistant drafts.

AC-FR-14.2b Confirm a draft with an invalid date and verify rejection under the ordinary date rule. Rules: BR-06, BR-10.

FR-14.3 The assistant shall state when permitted records or approved policy cannot support an answer.

AC-FR-14.3 Ask about an unprovided policy and verify an insufficient-information response. Rules: BR-11, BR-12.

FR-14.3a The system shall offer the relevant manual workflow when the assistant fails.

AC-FR-14.3a Disable the assistant and follow its fallback to complete the ordinary request form. Rules: BR-11, BR-12.

4.15 Sustainability and financial indicators

4.15.1 Description and priority

Calculate reproducible financial and sustainability indicators from qualified evidence.

Priority: Medium / P1. Scope: Essential. This completes required intelligence or reporting after trustworthy records and manual workflows exist. Source: S1 pp 6 and 8; S2 FR-15; S3 sections 5.3 and 11.4. Detailed use cases: UC-34, UC-35, UC-36.

4.15.2 Stimulus and response sequences

Actor action or stimulus	System response
Finance or sustainability officer reviews evidence	Qualify source records and reject duplicated or unsupported benefit entries.
Viewer selects period and department	Compute the stated formulas with units, estimate labels and evidence coverage.

4.15.3 Functional requirements

FR-15.1 The system shall calculate completed reuse units using section 3.4.

AC-FR-15.1 Complete transfers of three and two units in the period and verify five reuse units. Rules: BR-12, BR-13.

FR-15.1a The system shall calculate the publication-cohort reuse rate using section 3.4.

AC-FR-15.1a Receive four of ten first-published cohort units and verify 40 percent, without recounting later movements of those units. Rules: BR-12, BR-13.

FR-15.1b The system shall count repair completions using section 3.4.

AC-FR-15.1b Provide two completed repair tasks and one inspection-only task and verify two repair completions. Rules: BR-12, BR-13.

FR-15.1c The system shall calculate avoided purchase value using section 3.4.

AC-FR-15.1c Apply the qualified 20-chair fixture at AED 150 and verify AED 3000. Rules: BR-12, BR-13.

FR-15.1d The system shall calculate net reuse benefit using section 3.4.

AC-FR-15.1d Deduct AED 400 transfer and AED 300 repair from AED 3000 qualified value and verify AED 2300. Rules: BR-12, BR-13.

FR-15.1e The system shall calculate waste diversion using section 3.4.

AC-FR-15.1e Apply the otherwise-discarded 20-chair fixture at 8 kg each and verify 160 kg. Rules: BR-12, BR-13.

FR-15.1f The system shall calculate estimated life extension using section 3.4.

AC-FR-15.1f Apply a 365.25-day extension and the 30.4375-day monthly convention and verify 12 months labelled Estimated. Rules: BR-12, BR-13.

FR-15.1g The system shall calculate estimated avoided emissions using the evidence boundary and factors in section 3.4.

AC-FR-15.1g Run the desktop fixture in section 3.4.3 and verify 219.94 kg CO2e to two decimals. Rules: BR-12, BR-13.

FR-15.2 The system shall require finance-reviewed purchasing-need, quantity and replacement-price evidence before counting avoided purchase value.

AC-FR-15.2 Remove purchasing-need evidence from a completed reuse fixture and verify exclusion from confirmed avoided purchases. Rules: BR-13.

FR-15.2a The system shall deduct attributed transfer and repair costs when calculating net reuse benefit.

AC-FR-15.2a Apply the 20-chair cost fixture and verify AED 2300 net benefit. Rules: BR-13.

FR-15.3 The system shall link each benefit entry to a completed event and its evidence identifier.

AC-FR-15.3 Retrieve a benefit entry and verify both references resolve to its qualifying event and evidence. Rules: BR-02, BR-13.

FR-15.3a The system shall prevent duplicate counting of the same purchasing need or disposal-diversion episode.

AC-FR-15.3a Process the same qualified episode twice and verify that its benefit is counted once. Rules: BR-02, BR-13.

FR-15.3b The system shall retain an auditable benefit-correction history.

AC-FR-15.3b Correct one benefit assumption and verify the prior calculation, reason and reviewer remain retrievable. Rules: BR-02, BR-13.

FR-15.4 The system shall display Unknown instead of zero for indicators with missing required inputs.

AC-FR-15.4 Remove a required carbon factor and verify Unknown carbon while known financial values remain available. Rules: BR-12, BR-13.

FR-15.4a The system shall display evidence coverage for each indicator.

AC-FR-15.4a Provide ten eligible records with six complete evidence sets and verify coverage of 6/10. Rules: BR-12, BR-13.

FR-15.4b The system shall reject AI-generated values as carbon-factor evidence.

AC-FR-15.4b Submit a model-only factor with no approved source and verify it cannot qualify an emissions estimate. Rules: BR-12, BR-13.

FR-15.5 The system shall allow finance staff to review cost evidence with reviewer and time recorded.

AC-FR-15.5 Review a cost fixture as finance staff and verify the recorded decision, reviewer and timestamp. Rules: BR-01, BR-02.

FR-15.5a The system shall allow sustainability staff to review mass, factors, boundaries and life-extension evidence with reviewer and time recorded.

AC-FR-15.5a Review a sustainability fixture and verify the saved evidence decision and review metadata. Rules: BR-01, BR-02.

FR-15.5b The system shall prevent benefit-review actions from modifying completed workflow records.

AC-FR-15.5b Attempt to edit transfer history through the benefit dashboard and verify rejection. Rules: BR-01, BR-02.

4.16 Notifications and auditability

4.16.1 Description and priority

Notify the right recipients and preserve inspectable application history.

Priority: High / P0. Scope: Essential. This establishes identity, trustworthy records or the controlled transaction workflow required by later features. Source: S1 pp 6 and 8; S2 FR-16 and NFR-07; S3 section 3.6. Detailed use cases: UC-03, UC-04, UC-05, UC-07, UC-12, UC-15, UC-16, UC-17, UC-21, UC-24, UC-39, UC-40, UC-41.

4.16.2 Stimulus and response sequences

Actor action or stimulus	System response
A valid business event is recorded	Create recipient-specific notifications and audit records.
User reads notifications or administrator searches audit	Recheck scope; maintain read state without altering business history.

4.16.3 Functional requirements

FR-16.1 The system shall create in-application notifications for allocation-request submission, approval or rejection, cancellation or expiry, dispatch, receipt dispute or completion, and assigned maintenance updates for the recipients in section 2.

AC-FR-16.1 Trigger every event once and confirm only the defined recipients receive a linked notification; an unauthorized recipient sees no restricted details. Rules: BR-01, BR-14.

FR-16.2 The system shall allow recipients to view their permitted notifications.

AC-FR-16.2 Open a recipient's notification list and verify it contains only their permitted notifications. Rules: BR-14.

FR-16.2a The system shall allow recipients to mark their notifications read.

AC-FR-16.2a Mark one notification read and verify its persisted read state. Rules: BR-14.

FR-16.2b The system shall preserve a completed business action when notification delivery fails.

AC-FR-16.2b Fail delivery during a valid approval and verify that approval remains committed. Rules: BR-14.

FR-16.2c The system shall expose a retryable delivery record after notification failure.

AC-FR-16.2c Force delivery failure and verify one pending record available for retry. Rules: BR-14.

FR-16.3 The system shall create append-only application audit records for important changes and denied privileged actions, including actor, UTC time, action, target, outcome and relevant before/after values without credentials.

AC-FR-16.3 Change a role, reject approval and confirm receipt; inspect required audit fields; an ordinary user cannot edit or delete an event. Rules: BR-02, BR-15.

FR-16.4 The system shall let assigned asset administrators filter permitted audit records by actor, target, date and action.

AC-FR-16.4 Apply a known filter fixture and verify the returned event IDs against its expected set. Rules: BR-01, BR-02.

FR-16.4a The system shall restrict departmental history access to permitted assets.

AC-FR-16.4a Request another department's restricted history and verify no protected events are returned. Rules: BR-01, BR-02.

4.17 Operational reports and AI summaries

4.17.1 Description and priority

Provide conventional reports and optional human-reviewed AI summaries.

Priority: Medium / P1. Scope: Essential. This completes required intelligence or reporting after trustworthy records and manual workflows exist. Source: S1 pp 4 and 8; S2 FR-17; S3 sections 3.6 and 12. Detailed use cases: UC-37, UC-38.

4.17.2 Stimulus and response sequences

Actor action or stimulus	System response
Viewer selects report and filters	Compute conventional rows and totals from authoritative records.
Viewer requests and reviews an AI summary	Save only reviewed wording with source/model records; conventional reports survive AI failure.

4.17.3 Functional requirements

FR-17.1 The system shall produce date- and department-filtered reports of available assets, open requests, approvals, completed transfers, maintenance, ownership changes and asset status within the viewer permission scope.

AC-FR-17.1 Generate each report for a fixture period and department; rows and totals match saved records and exclude restricted data. Rules: BR-01.

FR-17.2 The system shall produce AI-written summaries of exchanges, maintenance trends, high-demand categories, savings and sustainability using verified report data, labelled estimates and references to the report period.

AC-FR-17.2 Compare every numeric statement in a generated summary with the underlying report; unsupported claims are rejected or flagged for human revision. Rules: BR-10, BR-12.

FR-17.3 The system shall keep conventional reports available if AI summary generation fails.

AC-FR-17.3 Disable AI and verify that the conventional report still renders with correct figures. Rules: BR-11.

FR-17.3a The system shall record model metadata for each saved AI summary.

AC-FR-17.3a Save a summary and verify its provider/model/configuration reference is retrievable. Rules: BR-11.

FR-17.3b The system shall record human review for each saved AI summary.

AC-FR-17.3b Save a reviewed summary and verify its reviewer, time and review outcome. Rules: BR-11.

5 Other nonfunctional requirements

These requirements apply to the stated academic prototype and measured test conditions. The classroom figures in S10 illustrate testability; they do not override the Phase 2 project targets. Targets are acceptance obligations for future testing, not achieved results or production guarantees.

5.1 Performance requirements

Search response is measured from submission to rendered result at the declared load. The 15-second AI failure/response bound is specified in NFR-06 under resilience. Fix the dataset, environment and workloads before evaluation so results are comparable.

NFR-01 Performance. For 200 sample assets and five concurrent users, at least 90 of 100 ordinary searches shall render results within three seconds of submission under the recorded test environment.

AC-NFR-01 Run the fixed 100-query workload; record client-to-render durations, failures, host and connection. At least 90 successful searches meet three seconds.

5.2 Safety requirements

Safety takes precedence over match ranking and speed. Publication, final allocation, dispatch and disposition require the applicable human condition/clearance evidence under BR-04 and FR-03/07/08/09. No university safety certification is claimed; actual policy and authorized handling evidence remain D-10.

NFR-16 Every attempted allocation, dispatch or disposition shall fail closed when required condition or clearance is missing, expired or unsafe.

AC-NFR-16 Run the invalid-clearance fixture across each protected action and verify zero forbidden transitions. Rules: BR-04, BR-09, BR-10.

NFR-16a AI output alone shall never constitute safety clearance.

AC-NFR-16a Supply an AI statement of safety without human clearance and verify that the clearance gate remains closed. Rules: BR-04, BR-09, BR-10.

5.3 Security requirements

NFR-03 Hosted sessions and protected downloads shall use HTTPS.

AC-NFR-03 Inspect all hosted protected routes in the test fixture and verify HTTPS transport.

NFR-03a Server-side access checks shall enforce every permission-matrix action.

AC-NFR-03a Run one permitted and one denied case per matrix action, including direct URLs and assistant paths; no denied case exposes protected content or changes records.

NFR-04 Passwords shall use the framework password-hashing facility.

AC-NFR-04 Create a test account and verify that the stored password is a framework hash rather than plaintext.

NFR-04a Credentials and API keys shall be absent from browser responses, application logs and tracked repository content.

AC-NFR-04a Search the stated surfaces for seeded test secrets and verify zero occurrences.

NFR-13 The prototype shall restrict its personal-data fixtures to fictional identities. UAE PDPL, Federal Decree-Law No. 45 of 2021, is the privacy baseline for the controls below. [S15]

AC-NFR-13 Compare all seeded and imported identity fields with the approved fictional fixture manifest; pass only if every identity is fictional and no unapproved identity is present.

NFR-13a The external-AI payload builder shall transmit only allowlisted, approved asset fields without personal identifiers.

AC-NFR-13a Capture every supported AI task payload containing seeded names, emails, IDs and secrets in excluded fields; pass with zero excluded-field values transmitted.

NFR-13b The system shall restrict sample exports and backups to explicitly authorized accounts.

AC-NFR-13b Attempt each export and backup-access route using an unauthorized account; pass with zero data bytes disclosed.

NFR-13c The prototype retention process shall delete hosted fictional personal data and backup copies within 90 days after grading closes unless a documented course retention obligation applies.

AC-NFR-13c Run the retention job with a controlled grading-close date and a 90-day cutoff; verify deletion of expired fixtures and backup copies, with any retained exception tied to an owner, reason and expiry.

NFR-13d Real-personal-data use shall remain disabled until an institutional reviewer records the UAE PDPL applicability assessment, lawful processing basis, purpose, retention, data-subject-rights procedure and external-provider/cross-border assessment. [S15]

AC-NFR-13d Leave each approval field absent in turn and attempt enabling real-data use; every incomplete approval fixture is denied. A complete signed test record permits the gate in an isolated fictional-data test only.

5.4 Software quality attributes

Correctness, safety and authority take precedence over convenience or AI availability. Manual continuity, usable forms, recovery and reproducible setup make the prototype demonstrable without claiming a production service level. Cost remains an organizational constraint inherited from Phase 2.

NFR-02 Usability. At least four of five representative or role-playing test users shall complete assigned search, request and status-tracking tasks without facilitator assistance after the same 15-minute introduction.

AC-NFR-02 Record each task outcome and assistance; a user passes only if all three tasks succeed without assistance. At least four users pass.

NFR-05 Stock, reservation, approval and receipt transactions shall commit atomically while conserving quantity.

AC-NFR-05 Inject a failure inside each transaction type and verify rollback to the full original quantity/state snapshot.

NFR-05a Retried stock, reservation, approval and receipt actions shall be idempotent.

AC-NFR-05a Retry each successful fixture action and verify that its state and quantity effects occur once.

NFR-06 An interactive AI request shall return a result or explicit timeout/failure within 15 seconds.

AC-NFR-06 Delay the AI provider beyond 15 seconds and verify visible failure feedback by the 15-second bound.

NFR-06a AI failure shall not prevent ordinary search, manual requests, approvals or reports.

AC-NFR-06a Disable AI and complete the fixed manual-continuity scenario through these workflows without an AI dependency.

NFR-07 AI match quality. At least eight of ten held-out requests shall have at least one suitable asset among the first three semantic recommendations, evaluated against independently reviewed relevance labels.

AC-NFR-07 Use the ten requests excluded from tuning; two team reviewers judge mandatory suitability and usefulness before scoring. Report success count and failures.

NFR-08 Classification quality. At least 40 of 50 held-out labelled asset examples shall receive the correct broad-category suggestion under the agreed category set.

AC-NFR-08 Record all 50 labels, outputs, task model and errors. Count abstentions as not correct; do not tune on these evaluation examples.

NFR-09 AI review quality. All explanations and saved summaries in the acceptance fixture set shall cite or identify supporting permitted records, preserve uncertainty and pass human review for unsupported numeric claims.

AC-NFR-09 Review 20 explanation/summary outputs; any fabricated number, permission leak or false approval is a failure regardless of average model quality.

NFR-10 A protected database backup shall be produced at least every 24 hours during active prototype use.

AC-NFR-10 Inspect the active-use backup schedule and timestamps and verify no interval exceeds 24 hours.

NFR-10a A restore shall recover the selected backup consistently into a separate test environment.

AC-NFR-10a Restore one dated backup and verify matching record counts, quantities, links and access controls.

NFR-11 The browser application shall conform to WCAG 2.1 Level AA across every delivered page and complete user process. [S14]

AC-NFR-11 Audit all pages, states and complete processes against every WCAG 2.1 Level A and AA success criterion using automated checks plus manual keyboard, screen-reader, contrast, zoom/reflow and error-handling checks. Pass only with zero applicable failures and a recorded justification for every not-applicable criterion; retain criterion ID, page/state, method and evidence. Automated results alone do not establish conformance.

NFR-12 Core workflow, data-access and AI-integration responsibilities shall be separated and documented.

AC-NFR-12 Review the module responsibility map and verify distinct documented boundaries for all three responsibilities.

NFR-12a Another team member shall be able to initialize sample data and run the project and checks using the README in a clean environment.

AC-NFR-12a Have a reviewer follow only the documented setup sequence and record successful initialization and execution.

NFR-14 Prototype cash spending shall remain within AED 450 unless the team records an approved revised limit.

AC-NFR-14 Reconcile the spending ledger against the applicable documented limit and verify no excess.

NFR-14a The application shall block optional AI calls at the configured allowance.

AC-NFR-14a Exhaust the application quota and verify that the next optional AI call is blocked before provider invocation.

NFR-14b The application shall alert at 80 percent of the configured AI allowance.

AC-NFR-14b Advance usage across the 80-percent threshold and verify the alert.

NFR-15 Report counts shall equal the underlying qualified fixture records exactly.

AC-NFR-15 Reconcile the report against 20 transfer and 20 maintenance fixtures, including correction and duplicate-event cases; every count matches.

NFR-15a Monetary outputs shall round final AED totals to two decimals without intermediate rounding.

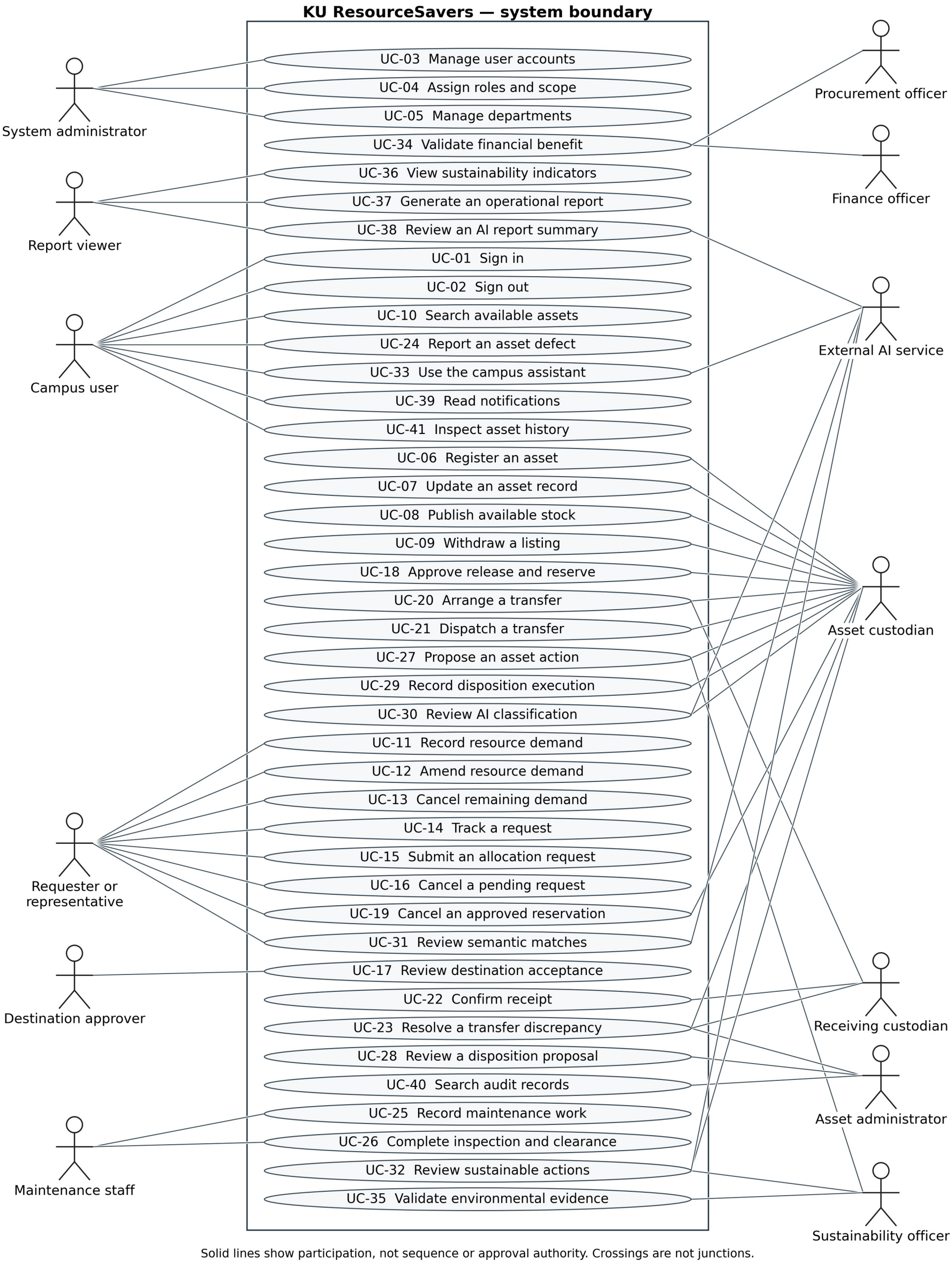
AC-NFR-15a Calculate a fixture containing fractional intermediate values and verify its final two-decimal total against an independent calculation.

NFR-17 Department names alone shall not change candidate eligibility, compatibility scores or ordering when permissions, locations, asset/request facts, approved factors and configuration are identical.

AC-NFR-17 Evaluate five paired requests differing only in neutral department names with fixed model outputs and identical other inputs, three times each. Pass only if all fifteen paired comparisons retain identical eligible IDs, scores and ordering; correct and retest every unexplained difference. Record live-model variation separately for investigation. This controlled check does not establish campus-wide fairness. Rules: BR-01, BR-10, BR-12.

6 Complete use case diagram

Figure 1 shows all 41 use cases and 13 distinct actors within one complete overview. Each actor appears once outside the system boundary. Every primary and supporting association matches section 7 and Appendix B. Use cases are grouped by primary actor for readability; position does not indicate execution order. Solid lines show participation, not automatic permission or approval authority.



Solid lines show participation, not sequence or approval authority. Crossings are not junctions.
Each actor and use case appears once. Roles, authority and flows are defined in section 7 and Appendix B.

Figure 1. Complete use-case overview: 13 actors and 41 use cases (UC-01–UC-41).

7 Detailed use case descriptions

Each use case follows the lecturer template: name, description, primary/supporting actors, preconditions, trigger, main flow, separate alternatives and exceptions, success/failure outcomes and rules. A1 or E1 identifies a branch at the stated main-flow step. Preconditions are rechecked when saving an authoritative change. Generic Campus user actions still require the action-specific permissions in section 2.

UC 01 Sign in

Description and goal Obtain an authenticated session for an active prototype account.
Primary actor Campus user
Supporting actors None
Preconditions An administrator-created active account exists; the user is not signed in.
Trigger The user selects Sign in.
Main flow
1. The user enters their credentials.
2. The system verifies the credentials and active-account status.
3. The system creates a session and loads only permitted navigation.
4. The user reaches their authorized starting page.

Alternative flows
- A1 at step 2: Incorrect credentials produce a neutral error; the user may correct them and return to step 1.

Exception flows
- E1 at step 2: Authentication service failure reports temporary unavailability and creates no session.

Success postconditions An authenticated session exists with the account permissions.
Failure postconditions No authenticated session is created and no protected data is exposed.
Rules BR-01. See section 2.6 for rule definitions.
Requirement links **FR-01.2, FR-01.2a, FR-01.3, NFR-03, NFR-03a, NFR-04, NFR-04a, NFR-11**

UC 02 Sign out

Description and goal End access through the current authenticated session.
Primary actor Campus user
Supporting actors None
Preconditions The user has an authenticated session.
Trigger The user selects Sign out.
Main flow
1. The user confirms sign-out if unsaved input needs a warning.
2. The system invalidates the current session.
3. The system removes protected page content from the active view.
4. The system displays the sign-in page.

Alternative flows
- A1 at step 1: The user stays to finish an unsaved form; the session remains active and the use case ends.

Exception flows
- E1 at step 2: If invalidation cannot be confirmed, report the failure and do not falsely claim that sign-out succeeded.

Success postconditions The old session cannot perform another protected action.
Failure postconditions No successful sign-out message is shown without invalidation.
Rules BR-01. See section 2.6 for rule definitions.
Requirement links **FR-01.2, FR-01.2a, NFR-03, NFR-03a**

UC 03 Manage user accounts

Description and goal Create or deactivate accounts while retaining historical actor references.
Primary actor System administrator
Supporting actors None
Preconditions The administrator is signed in with account-management permission.
Trigger The administrator opens account management.
Main flow
1. The administrator chooses Create or Deactivate and supplies the account details.
2. The system validates identity uniqueness and the active department reference.
3. The administrator confirms the change.
4. The system saves the account change and audit event and applies access status to subsequent actions.

Alternative flows
- A1 at step 2: A duplicate identity is identified; correct it and return to step 1.

Exception flows
- E1 at step 4: Saving fails; retain the original account state and show a retryable error.

Success postconditions The account state is recorded without deleting its historical references.
Failure postconditions No partial account update or unauthorized access change occurs.
Rules BR-01, BR-02. See section 2.6 for rule definitions.
Requirement links FR-01.1, FR-01.1a, FR-01.1b, FR-01.1c, FR-01.4, FR-01.4a, FR-16.3, NFR-04, NFR-04a, NFR-10, NFR-10a, NFR-12, NFR-12a

UC 04 Assign roles and scope

Description and goal Grant or remove explicit roles and departmental scope.
Primary actor System administrator
Supporting actors None
Preconditions The administrator and target account exist; the administrator may manage role assignments.
Trigger The administrator selects the target account roles.
Main flow
1. The system displays existing assignments.
2. The administrator selects roles and departments.
3. The system validates each role and scope and shows the proposed change.
4. The administrator confirms; the system saves and audits the assignments.

Alternative flows
- A1 at step 3: A department is inactive; choose an active department and return to step 2.

Exception flows
- E1 at step 4: Permission is revoked or saving fails; reject the update and retain previous assignments.

Success postconditions Explicit assignments govern subsequent actions; technical administration grants no implicit business approval.
Failure postconditions No partial role change is stored.
Rules BR-01, BR-08. See section 2.6 for rule definitions.
Requirement links FR-01.1, FR-01.1a, FR-01.1b, FR-01.1c, FR-01.3, FR-16.3, NFR-03, NFR-03a

UC 05 Manage departments

Description and goal Maintain the department list without orphaning assets or workflows.
Primary actor System administrator
Supporting actors None
Preconditions The administrator has department-management permission.
Trigger The administrator opens a department record or chooses Create.
Main flow
1. The administrator enters a name or proposes deactivation.
2. The system validates uniqueness and unresolved holdings or workflows.
3. The system displays any restrictions.
4. The administrator confirms an eligible change and the system records it with history.

Alternative flows
- A1 at step 2: Unresolved holdings block deactivation; retain the active department and end without change.

Exception flows
- E1 at step 4: Saving fails; the existing department state is retained.

Success postconditions A valid department change is recorded and prior references remain usable.
Failure postconditions No orphaned reference or partial deactivation occurs.
Rules BR-01, BR-02. See section 2.6 for rule definitions.
Requirement links FR-01.4, FR-01.4a, FR-16.3

UC 06 Register an asset

Description and goal Create a traceable individual asset or homogeneous lot.
Primary actor Asset custodian
Supporting actors None
Preconditions The custodian may register assets for an active department.
Trigger The custodian selects Register asset.
Main flow
1. The system presents required fields and attachment limits.
2. The custodian enters identity, condition, quantity, specifications and location.
3. The system validates the data and optionally offers UC-30 classification suggestions.
4. The custodian confirms; the system assigns a unique ID, records history and creates InUse or Blocked stock without publishing it.

Alternative flows
- A1 at step 3: Optional information is unknown; retain an explicit Unknown value and continue to step 4.
- A2 at step 3: The user invokes UC-30, reviews the suggestions and returns to validation.

Exception flows
- E1 at step 3: Invalid quantity, duplicate tag or forbidden attachment is rejected; correct the fields and retry step 2.
- E2 at step 4: Saving fails; no partial asset or attachment reference remains.

Success postconditions The asset has an immutable ID, assigned custody and conserved quantity.
Failure postconditions No partial registration or public listing is created.
Rules BR-01, BR-02, BR-03, BR-15. See section 2.6 for rule definitions.
Requirement links FR-02.1, FR-02.1a, FR-02.3, FR-02.3a, FR-02.3b, FR-02.3c, FR-02.3d, FR-02.4, FR-02.4a, FR-02.4b, FR-02.5, FR-02.5a, FR-02.5b, FR-02.5c, FR-08.1

UC 07 Update an asset record

Description and goal Correct permitted descriptive fields while preserving history.
Primary actor Asset custodian
Supporting actors None
Preconditions The asset exists and the custodian has assignment scope.
Trigger The custodian opens Edit asset.
Main flow
1. The system displays editable and workflow-controlled fields.
2. The custodian changes descriptive data or permitted attachments.
3. The system validates fields and blocks direct quantity or custody edits.
4. The custodian confirms; the system saves the new values and before/after history.

Alternative flows
- A1 at step 2: The custodian cancels editing; retain all saved values and end.

Exception flows
- E1 at step 3: A tag conflicts or an attachment is invalid; return to step 2 with field errors.
- E2 at step 4: Saving fails; retain the complete previous record.

Success postconditions The permitted update is stored with a traceable prior value.
Failure postconditions Workflow balances and original data remain unchanged.
Rules BR-01, BR-02, BR-03. See section 2.6 for rule definitions.
Requirement links FR-02.2, FR-02.2a, FR-02.2b, FR-02.3, FR-02.3a, FR-02.3b, FR-02.3c, FR-02.3d, FR-02.5, FR-02.5a, FR-02.5b, FR-02.5c, FR-16.3

UC 08 Publish available stock

Description and goal Make eligible surplus units discoverable for allocation requests.
Primary actor Asset custodian
Supporting actors None
Preconditions The assigned asset is Active with complete required fields and unoffered InUse units.
Trigger The custodian selects Publish quantity.
Main flow
1. The custodian enters the offered quantity.
2. The system checks current condition, specifications and InUse balance.
3. The system previews the eligible quantity to publish.
4. The custodian confirms; the system moves those units to Free, marks Published and records the event.

Alternative flows
- A1 at step 2: Offer fewer units than the available lot; validate the reduced quantity and continue to step 3.

Exception flows
- E1 at step 2: Unsafe condition or missing clearance prevents publication; retain stock state.
- E2 at step 4: Concurrent change or save failure rejects the operation without partial movement.

Success postconditions The offered eligible quantity appears in search as Free.
Failure postconditions No ineligible publication or partial quantity movement occurs.
Rules BR-03, BR-04. See section 2.6 for rule definitions.
Requirement links FR-03.1, FR-03.2, FR-03.2a, NFR-16, NFR-16a

UC 09 Withdraw a listing

Description and goal Remove unheld offered units from further allocation.
Primary actor Asset custodian
Supporting actors None
Preconditions The custodian controls a published listing with Free units.
Trigger The custodian selects Withdraw.
Main flow
1. The system displays Free quantity and existing approved commitments.
2. The custodian selects an unheld quantity and provides a reason.
3. The system validates the current Free balance.
4. The system moves the selected units to InUse and updates publication visibility and history.

Alternative flows
- A1 at step 2: Withdraw only part of the Free quantity; the remainder stays Published.
- A2 at step 2: A safety issue is reported through UC-24; affected stock follows BR-04.

Exception flows
- E1 at step 3: Requested quantity includes Held units; refuse and explain the cancellation route.
- E2 at step 4: Saving fails; keep previous publication and balances.

Success postconditions Withdrawn units cannot be newly requested or approved; existing approved holds remain traceable.
Failure postconditions No hold is silently cancelled or released.
Rules BR-03, BR-04, BR-05. See section 2.6 for rule definitions.
Requirement links **FR-03.2, FR-03.2a, FR-03.3, FR-03.3a, FR-03.3b**

UC 10 Search available assets

Description and goal Find published resources within the user permission scope.
Primary actor Campus user
Supporting actors None
Preconditions The user is signed in and allowed to view campus listings.
Trigger The user enters a category, keyword or filters.
Main flow
1. The user submits search criteria.
2. The system applies permission, publication and filter checks.
3. The system displays matching assets, Free quantities and known condition/location.
4. The user opens a permitted result or changes the criteria.

Alternative flows
- A1 at step 3: No matches exist; display guidance and return to step 1 with revised criteria.
- A2 at step 4: The user requests semantic matching through UC-31; ordinary search remains available.

Exception flows
- E1 at step 2: Search service failure shows an error and allows retry at step 1 without inventing results.

Success postconditions The user receives current permitted results; searching creates no request or hold.
Failure postconditions No protected data is disclosed and stock is unchanged.
Rules BR-01, BR-04, BR-11, BR-12. See section 2.6 for rule definitions.
Requirement links **FR-05.1, FR-05.1a, FR-05.2, FR-05.2a, FR-05.2b, FR-05.3, FR-05.3a, NFR-01, NFR-02**

UC 11 Record resource demand

Description and goal Record a departmental need that can be matched to one or more assets.
Primary actor Requester or representative
Supporting actors None
Preconditions The requester may create demand for the destination department.
Trigger The requester selects New resource demand.
Main flow
1. The requester enters category, purpose, quantity, specifications, condition, urgency, date and destination.
2. The system validates required fields and user scope.
3. The system shows the demand for confirmation.
4. The requester confirms; the system assigns an ID and creates Open demand with zero received quantity.

Alternative flows
- A1 at step 1: Use UC-33 to prepare a draft, review its fields and return to step 2.

Exception flows
- E1 at step 2: Invalid date or quantity is rejected with field errors; return to step 1.
- E2 at step 4: Saving fails and no demand record is created.

Success postconditions Open demand exists independently of any allocation request; no stock is held.
Failure postconditions No partial demand, allocation request or reservation exists.
Rules BR-01, BR-06, BR-15. See section 2.6 for rule definitions.
Requirement links FR-04.1, NFR-02, NFR-11

UC 12 Amend resource demand

Description and goal Correct a demand record before dependent activity makes the change unsafe.
Primary actor Requester or representative
Supporting actors None
Preconditions The requester can access an Open or PartiallyFulfilled demand record.
Trigger The requester selects Amend demand.
Main flow
1. The system displays editable fields and dependent records.
2. The requester proposes changed specifications, quantity, date or destination.
3. The system checks that no pending request or active commitment exists and that quantity is at least the received quantity.
4. The requester confirms; the system saves and audits the revision.

Alternative flows
- A1 at step 3: A pending request or approved hold blocks amendment; cancel it through UC-16 or UC-19 and restart at step 1.

Exception flows
- E1 at step 3: The record changes concurrently or becomes in transit; reject the revision.
- E2 at step 4: Saving fails; preserve the previous demand.

Success postconditions A valid demand revision is saved without changing historical receipts.
Failure postconditions No dependent request or transfer silently changes meaning.
Rules BR-02, BR-06. See section 2.6 for rule definitions.
Requirement links FR-04.2, FR-04.2a, FR-16.3

UC 13 Cancel remaining demand

Description and goal Close unneeded demand while preserving completed reuse.
Primary actor Requester or representative
Supporting actors None
Preconditions The requester has access to nonterminal demand with an outstanding remainder.
Trigger The requester selects Cancel remaining demand.
Main flow
1. The system displays receipts, pending requests and active commitments.
2. The requester confirms cancellation and a reason.
3. The system rejects cancellation if any transfer is InTransit or Disputed.
4. Otherwise the system cancels pending requests, releases approved pre-dispatch holds once and closes the remainder.

Alternative flows
- A1 at step 1: Some units are already received; preserve that quantity and cancel only the remainder.

Exception flows
- E1 at step 3: A transfer is in transit or disputed; explain its reference and retain demand.
- E2 at step 4: Saving fails; all linked states and quantities remain as before.

Success postconditions Outstanding demand is Cancelled; previous receipts and benefits remain traceable.
Failure postconditions No partial cancellation or silent in-transit stock release occurs.
Rules BR-02, BR-05, BR-06. See section 2.6 for rule definitions.
Requirement links **FR-04.4, FR-04.4a, FR-04.4b, FR-04.4c, FR-06.5, FR-06.5a, FR-06.5b, NFR-05, NFR-05a**

UC 14 Track a request

Description and goal Inspect demand, allocation decisions and receipt progress.
Primary actor Requester or representative
Supporting actors None
Preconditions The requester has permitted demand or allocation records.
Trigger The requester opens request status.
Main flow
1. The requester selects a demand or allocation request ID.
2. The system checks record scope and retrieves linked records.
3. The system displays pending decisions, approved commitments, dispatch and received quantities separately.
4. The requester follows a permitted transfer, decision or notification link.

Alternative flows
- A1 at step 3: Partial receipt exists; show PartiallyFulfilled demand and the remaining quantity.

Exception flows
- E1 at step 2: Access is denied or the record is absent; disclose no restricted details.
- E2 at step 2: Retrieval fails; show a retry action without stale information presented as current.

Success postconditions The requester can distinguish submission, approval, dispatch and fulfilment.
Failure postconditions No workflow state or quantity changes during viewing.
Rules BR-01, BR-06, BR-07. See section 2.6 for rule definitions.
Requirement links **FR-04.3, FR-04.3a, FR-06.3, FR-06.3a, FR-07.5, NFR-02**

UC 15 Submit an allocation request

Description and goal Ask for a specified asset quantity without reserving it at submission.

Primary actor Requester or representative

Supporting actors None

Preconditions Open demand exists; the requester can view a published eligible asset with sufficient current Free stock.

Trigger The requester selects Request this asset.

Main flow

1. The requester selects demand, asset and quantity and confirms purpose, destination and required date.
2. The system checks positive whole quantity, current Free stock, uncommitted demand and eligibility.
3. The requester confirms submission.
4. The system assigns an allocation request ID and 48-hour pending expiry, records PendingApproval and notifies the destination approver without moving stock.

Alternative flows

- A1 at step 2: Quantity exceeds current availability; revise it and return to step 1.
- A2 at step 1: No demand exists yet; complete UC-11 and return to this step.

Exception flows

- E1 at step 2: The asset becomes unpublished or unsafe; refuse submission and offer a new search.
- E2 at step 4: Saving fails; no allocation request or hold is created.

Success postconditions A non-binding PendingApproval request exists; ownership, location and all stock buckets are unchanged.

Failure postconditions No partial request, approval or reservation is stored.

Rules BR-03, BR-04, BR-06, BR-14, BR-15. See section 2.6 for rule definitions.

Requirement links FR-06.1, FR-06.1a, FR-06.3, FR-06.3a, FR-06.4, FR-06.4a, FR-06.4b, FR-16.1, NFR-11

UC 16 Cancel a pending request

Description and goal Withdraw an allocation request before final approval.

Primary actor Requester or representative

Supporting actors None

Preconditions The requester controls a PendingApproval allocation request.

Trigger The requester selects Cancel allocation request.

Main flow

1. The system displays the request and any destination decision.
2. The requester confirms cancellation and reason.
3. The system rechecks that final approval has not occurred.
4. The system records Cancelled, preserves prior decisions and notifies participants without stock movement.

Alternative flows

- A1 at step 3: The pending window has expired; show Expired and end with no stock movement.

Exception flows

- E1 at step 3: Final approval already succeeded; refuse this route and offer UC-19.
- E2 at step 4: Saving fails; retain the pending state and report failure.

Success postconditions The request cannot be approved; there is no stock release because no hold existed.

Failure postconditions The request and balances remain consistent.

Rules BR-02, BR-05, BR-14. See section 2.6 for rule definitions.

Requirement links FR-06.4, FR-06.4a, FR-06.4b, FR-06.5, FR-06.5a, FR-06.5b, FR-16.1

UC 17 Review destination acceptance

Description and goal Accept or reject the department need before source release review.

Primary actor Destination approver

Supporting actors None

Preconditions The approver has destination scope, is not the requester, and the allocation request is unexpired and pending.

Trigger The approver opens the destination review queue.

Main flow

1. The system displays purpose, quantity, destination and asset details.
2. The approver reviews suitability and departmental need.
3. The approver accepts or rejects; rejection includes a reason.
4. The system revalidates authority, records decision and time, and routes acceptance to the source custodian with no stock movement.

Alternative flows

- A1 at step 3: Reject the request; the system records Rejected, notifies participants and creates no reservation.

Exception flows

- E1 at step 4: Expiry or revoked authority prevents the decision.
- E2 at step 4: Saving fails; no destination decision or stock change is stored.

Success postconditions Acceptance enables source review, or rejection closes the request; no hold exists yet.

Failure postconditions No approval is represented as saved when recording fails.

Rules BR-01, BR-05, BR-08, BR-14, BR-15. See section 2.6 for rule definitions.

Requirement links FR-07.1, FR-07.1a, FR-07.1b, FR-07.1c, FR-07.2, FR-07.2a, FR-07.2b, FR-16.1

UC 18 Approve release and reserve

Description and goal Authorize source release and reserve stock in the same successful transaction.

Primary actor Asset custodian

Supporting actors None

Preconditions Destination acceptance exists; the source custodian is assigned, distinct from the destination approver and requester, and the request is still pending.

Trigger The custodian opens source release review.

Main flow

1. The system displays the request, acceptance and current stock.
2. The custodian selects Approve release.
3. The system rechecks authority, expiry, publication, safety, Free stock and uncommitted demand.
4. The system records final approval, creates one linked Approved reservation, moves Free to Held and records notification delivery as pending in one business change.

Alternative flows

- A1 at step 2: Reject with a reason; record Rejected and no hold.
- A2 at step 3: Stock or demand is insufficient; keep the request pending, show the conflict and allow later retry before expiry or cancellation.

Exception flows

- E1 at step 3: Missing acceptance, self-approval or safety restriction denies the operation.
- E2 at step 4: Saving fails; request stays pending, no reservation is created and all stock remains unchanged.

Success postconditions One approved hold exists; custody and location remain with the source.

Failure postconditions No partial approval, duplicate hold or over-allocation occurs.

Rules BR-01, BR-03, BR-04, BR-05, BR-06, BR-08, BR-14. See section 2.6 for rule definitions.

Requirement links FR-06.2, FR-06.2a, FR-06.3, FR-06.3a, FR-07.1, FR-07.1a, FR-07.1b, FR-07.1c, FR-07.2, FR-07.2a, FR-07.2b, NFR-05, NFR-05a, NFR-16, NFR-16a

UC 19 Cancel an approved reservation

Description and goal Release an approved hold before dispatch.
Primary actor Requester or representative
Supporting actors Asset custodian
Preconditions An Approved or Converted reservation exists with no InTransit or Disputed transfer.
Trigger The requester or assigned source custodian opens pre-dispatch cancellation.
Main flow
1. The system displays the reservation and any prepared transfer.
2. The actor confirms cancellation with a reason.
3. The system rechecks that dispatch has not occurred.
4. The system cancels the reservation and prepared transfer, releases the held quantity once according to BR-05 and notifies participants.

Alternative flows
- A1 at step 4: The asset is unsafe; release the cancelled Held quantity into Blocked rather than Free.

Exception flows
- E1 at step 3: Dispatch has occurred; deny cancellation and preserve transit custody.
- E2 at step 4: Saving fails; retain the hold and linked states.

Success postconditions The hold no longer commits demand and released units occupy the correct bucket.
Failure postconditions No double release or partial linked cancellation occurs.
Rules BR-03, BR-04, BR-05, BR-06, BR-14. See section 2.6 for rule definitions.
Requirement links **FR-06.5, FR-06.5a, FR-06.5b, FR-07.3, FR-07.3a, FR-07.3b, NFR-05, NFR-05a**

UC 20 Arrange a transfer

Description and goal Prepare a handover from an approved reservation.
Primary actor Asset custodian
Supporting actors Receiving custodian
Preconditions The custodian controls an Approved reservation with both approval decisions.
Trigger The custodian selects Create transfer.
Main flow
1. The system displays the approved quantity and destinations.
2. The custodian enters receiving custodian, collection or delivery details and required evidence.
3. The system validates active assignments and required fields.
4. The custodian confirms; the system creates one Approved transfer linked to the reservation and marks the reservation Converted without another stock deduction.

Alternative flows
- A1 at step 4: A transfer already exists for this reservation; open that record instead of creating another.

Exception flows
- E1 at step 3: Receiver or evidence is invalid; return to step 2.
- E2 at step 4: Saving fails; the approved reservation remains responsible for its hold.

Success postconditions One transfer controls the existing hold with traceable arrangements.
Failure postconditions Held quantity is never deducted twice.
Rules BR-01, BR-03, BR-07. See section 2.6 for rule definitions.
Requirement links **FR-07.3, FR-07.3a, FR-07.3b, FR-06.3, FR-06.3a**

UC 21 Dispatch a transfer

Description and goal Record actual departure after all release conditions pass.
Primary actor Asset custodian
Supporting actors None
Preconditions An Approved transfer and corresponding Held quantity exist.
Trigger The source custodian selects Record dispatch.
Main flow
1. The custodian confirms the actual quantity and handover details.
2. The system rechecks approvals, scope and current safety eligibility.
3. The custodian confirms dispatch.
4. The system moves Held to InTransit, records departure time and notifies the receiver; source custody remains until receipt.

Alternative flows
- A1 at step 3: The custodian postpones dispatch; retain Approved and Held.

Exception flows
- E1 at step 2: A safety issue or missing valid approval blocks dispatch.
- E2 at step 4: Saving fails; transfer and Held quantity remain unchanged.

Success postconditions The transfer is InTransit and received demand is unchanged.
Failure postconditions No partial departure or duplicate movement is recorded.
Rules BR-03, BR-04, BR-07, BR-14, BR-15. See section 2.6 for rule definitions.
Requirement links **FR-07.4, FR-07.4a, FR-07.4b, FR-16.1, NFR-05, NFR-05a, NFR-16, NFR-16a**

UC 22 Confirm receipt

Description and goal Confirm delivery and update destination custody and fulfilment.
Primary actor Receiving custodian
Supporting actors None
Preconditions The transfer is InTransit and the receiver is assigned to its destination.
Trigger The receiver opens Record receipt.
Main flow
1. The receiver compares delivered units and condition against the dispatched record.
2. The receiver confirms a complete acceptable delivery.
3. The system rechecks scope and the unresolved transfer.
4. The system records Completed, creates destination InUse holdings, updates demand received quantity and emits history and notifications once.

Alternative flows
- A1 at step 2: Quantity or condition differs; record Disputed without benefit or ordinary completion and route to UC-23.

Exception flows
- E1 at step 3: Permission or transfer state changed; refuse the receipt.
- E2 at step 4: Saving fails; retain the entire transfer InTransit without partial fulfilment.

Success postconditions Completed receipt changes departmental custody/location; legal university ownership stays unchanged.
Failure postconditions No duplicate receipt, unaccounted quantity or premature benefit occurs.
Rules BR-02, BR-03, BR-06, BR-07, BR-13. See section 2.6 for rule definitions.
Requirement links **FR-07.5, FR-07.6, FR-07.6a, FR-07.6b, FR-02.4, FR-02.4a, FR-02.4b, FR-04.3, FR-04.3a, NFR-05, NFR-05a, NFR-11, NFR-10, NFR-10a**

UC 23 Resolve a transfer discrepancy

Description and goal Account for every disputed unit through evidenced acceptance or complete return.

Primary actor Asset administrator

Supporting actors Asset custodian, Receiving custodian

Preconditions The transfer is Disputed and the administrator has resolution authority.

Trigger The administrator opens the dispute record.

Main flow

1. The system displays dispatch, receipt observations and supporting evidence.
2. The administrator obtains source/receiver evidence and proposes full acceptance or complete return.
3. The relevant custodian confirms the actual outcome.
4. The system validates quantity conservation and records resolution, corrected custody/condition and the appropriate demand result.

Alternative flows

- A1 at step 2: Accept all units with documented damage; complete receipt with damaged holdings Blocked.
- A2 at step 2: Confirm full return; restore source InUse or Blocked holdings, record Returned and release the demand commitment.

Exception flows

- E1 at step 3: Missing units or insufficient evidence keep the dispute open and excluded from benefits.
- E2 at step 4: Saving fails; no partial resolution occurs.

Success postconditions Every unit is accounted for with an auditable resolution.

Failure postconditions Unexplained losses remain Disputed; no silent write-off occurs.

Rules BR-02, BR-03, BR-06, BR-07, BR-09, BR-13. See section 2.6 for rule definitions.

Requirement links FR-07.6, FR-07.6a, FR-07.6b, FR-08.1, NFR-05, NFR-05a

UC 24 Report an asset defect

Description and goal Create an inspection task and enforce any immediate safety restriction.

Primary actor Campus user

Supporting actors None

Preconditions The reporter may access the asset.

Trigger The reporter selects Report defect.

Main flow

1. The reporter enters symptoms and whether there is a safety concern.
2. The system validates the asset and report.
3. The reporter confirms submission.
4. The system creates a Reported task, applies safety blocking where required and notifies the custodian and assigned maintenance staff.

Alternative flows

- A1 at step 4: Approved holds already exist; retain their quantities but suspend dispatch pending clearance or cancellation.

Exception flows

- E1 at step 2: The asset is outside scope; deny the report without disclosing details.
- E2 at step 4: Saving fails; clearly report failure and do not claim that an inspection task exists.

Success postconditions The defect is recorded and required allocation/dispatch restrictions apply.

Failure postconditions No incomplete task is represented as successfully recorded.

Rules BR-01, BR-04, BR-14. See section 2.6 for rule definitions.

Requirement links FR-09.1, FR-09.1a, FR-09.1b, FR-09.1c, FR-16.1, NFR-16, NFR-16a

UC 25 Record maintenance work

Description and goal Document inspection, service actions and costs for an assigned task.
Primary actor Maintenance staff
Supporting actors None
Preconditions The staff member has an assigned Reported, UnderInspection or InProgress task.
Trigger The staff member opens the task.
Main flow
1. The staff member starts inspection or resumes work.
2. The system validates assignment and permitted transition.
3. The staff member records findings, actions, parts, costs and service dates.
4. The system validates values, preserves prior entries and saves UnderInspection or InProgress status.

Alternative flows
- A1 at step 3: Repair is uneconomical; record the finding and route a proposed next action to UC-27.
- A2 at step 1: A duplicate task is cancelled with a reason; this alone never clears a safety restriction.

Exception flows
- E1 at step 4: Invalid dates or negative costs require correction at step 3.
- E2 at step 4: Saving fails; previously recorded work remains intact.

Success postconditions Work and cost evidence are recorded without implying final clearance.
Failure postconditions No unauthorized maintenance or silent safety clearance occurs.
Rules BR-01, BR-02, BR-04, BR-09. See section 2.6 for rule definitions.
Requirement links FR-09.2, FR-09.2a, FR-09.4, FR-08.1

UC 26 Complete inspection and clearance

Description and goal Record final condition and decide whether an asset is fit for use.
Primary actor Maintenance staff
Supporting actors None
Preconditions The assigned task is UnderInspection or InProgress and findings are available.
Trigger The staff member selects Complete task.
Main flow
1. The staff member records final condition and an explicit fit-for-use decision.
2. The system checks required findings and clearance evidence.
3. The staff member confirms completion.
4. The system records Completed; eligible unheld Blocked units return to InUse, while others remain Blocked until resolved.

Alternative flows
- A1 at step 1: The asset is not fit for use; complete the work record with a negative decision and retain Blocked stock.
- A2 at step 4: Approved Held units exist; retain Held and clear only the documented safety restriction.

Exception flows
- E1 at step 2: Missing condition or evidence prevents closure.
- E2 at step 4: Saving fails; the previous task and safety states remain unchanged.

Success postconditions The clearance decision is auditable; republication requires UC-08.
Failure postconditions No unsafe stock becomes Free automatically.
Rules BR-03, BR-04, BR-09. See section 2.6 for rule definitions.
Requirement links FR-09.3, FR-09.3a, FR-09.4, NFR-16, NFR-16a

UC 27 Propose an asset action

Description and goal Request a documented retention, recovery or disposition decision.
Primary actor Asset custodian
Supporting actors Sustainability officer
Preconditions The proposer can access the asset and supporting evidence.
Trigger The proposer selects Propose action.
Main flow
1. The proposer selects retention, repair, refurbishment, donation, parts recovery, recycling, retirement or disposal.
2. The system displays condition, commitments and known supporting evidence.
3. The proposer enters quantity, reason and evidence.
4. The system validates and records Proposed without changing stock.

Alternative flows
- A1 at step 1: Use UC-32 for a suggestion, review it and return to the proposal.
- A2 at step 1: Repair or refurbishment is selected; route actual work through UC-25 and UC-26.

Exception flows
- E1 at step 3: Missing reason or invalid quantity requires correction.
- E2 at step 4: Saving fails; no proposal or disposition movement occurs.

Success postconditions A human proposal exists with evidence and no executed outcome.
Failure postconditions Advice and proposal alone never change custody or stock.
Rules BR-04, BR-09, BR-10, BR-12. See section 2.6 for rule definitions.
Requirement links **FR-08.2, FR-09.4**

UC 28 Review a disposition proposal

Description and goal Approve or reject consequential disposition independently of its proposer.
Primary actor Asset administrator
Supporting actors None
Preconditions A Proposed donation, recovery, recycling, retirement or disposal record exists; the administrator is distinct from its proposer.
Trigger The administrator opens the proposal.
Main flow
1. The system displays alternatives, condition, quantity and evidence.
2. The administrator reviews safety, authority and unresolved commitments.
3. The administrator approves or rejects with a reason.
4. The system rechecks scope, distinct identity and eligibility and records the decision without executing it.

Alternative flows
- A1 at step 3: Reject because reuse remains suitable; record the reason and notify the proposer.

Exception flows
- E1 at step 4: Active holds, missing clearance or self-approval block authorization.
- E2 at step 4: Saving fails; proposal remains Proposed.

Success postconditions An authorized decision is recorded; physical execution remains separate.
Failure postconditions No unauthorized approval or retirement movement occurs.
Rules BR-01, BR-04, BR-08, BR-09. See section 2.6 for rule definitions.
Requirement links **FR-08.3, NFR-16, NFR-16a**

UC 29 Record disposition execution

Description and goal Record an actually completed authorized action and retain its history.
Primary actor Asset custodian
Supporting actors None
Preconditions A valid approved disposition exists for units with no unresolved commitments.
Trigger The custodian selects Record execution.
Main flow
1. The custodian enters executed quantity, date, destination or handler and evidence.
2. The system rechecks approval, safety and current quantity.
3. The custodian confirms the actual event.
4. The system marks Completed, moves the affected units to Retired and records the specific donated/recycled/disposed life-cycle outcome.

Alternative flows
- A1 at step 3: Execution has not occurred; cancel recording and retain Approved without stock change.

Exception flows
- E1 at step 2: Approval, clearance or quantity is invalid; refuse execution recording.
- E2 at step 4: Saving fails or a duplicate submission arrives; preserve the original result without a second movement.

Success postconditions Executed disposition is traceable and quantity is conserved historically.
Failure postconditions No action is claimed as executed merely because it was approved.
Rules BR-02, BR-03, BR-04, BR-09, BR-15. See section 2.6 for rule definitions.
Requirement links **FR-08.4, FR-08.4a, FR-08.4b, FR-08.1, NFR-16, NFR-16a, NFR-10, NFR-10a**

UC 30 Review AI classification

Description and goal Use AI suggestions to improve asset metadata under custodian control.
Primary actor Asset custodian
Supporting actors External AI service
Preconditions The custodian is entering or editing a permitted asset with supported input.
Trigger The custodian selects Suggest classification.
Main flow
1. The system sends only approved minimized input to the AI service.
2. The service returns category, tags and other supported suggestions with metadata.
3. The system displays reasons, limitations and available confidence separately from saved values.
4. The custodian accepts, edits or rejects suggestions before normal validation and saving through UC-06 or UC-07.

Alternative flows
- A1 at step 4: Reject or correct a suggestion; preserve the user choice and record the override.

Exception flows
- E1 at step 2: Timeout, unreadable input or missing evidence produces an explicit failure and manual-entry route.
- E2 at step 2: Embedded instructions request a policy override; treat them as data and do not change authority.

Success postconditions Only user-confirmed, validated fields become saved data.
Failure postconditions No silent mutation or loss of manual input occurs.
Rules BR-01, BR-10, BR-11, BR-12. See section 2.6 for rule definitions.
Requirement links **FR-10.1, FR-10.2, FR-10.2a, FR-10.3, FR-10.3a, FR-12.1, FR-12.1a, FR-12.1b, FR-12.2, FR-12.2a, FR-12.3, FR-12.4, FR-12.4a, NFR-06, NFR-06a, NFR-08, NFR-09, NFR-13, NFR-13a, NFR-13b, NFR-13c, NFR-13d**

UC 31 Review semantic matches

Description and goal Find and compare suitable assets for recorded resource demand.

Primary actor Requester or representative

Supporting actors External AI service

Preconditions The requester can access Open or PartiallyFulfilled demand.

Trigger The requester selects Find AI matches.

Main flow

1. The system excludes unauthorized, unsafe, unavailable and mandatory-incompatible assets.
2. The service compares meaning and ranks eligible candidates using documented factors.
3. The system displays up to three candidates, reasons, ranking score, uncertainty and model metadata.
4. The requester compares candidates with ordinary search and may start UC-15 for a selected asset.

Alternative flows

- A1 at step 3: Fewer than three eligible assets exist; show only those assets.
- A2 at step 4: Reject the ranking and use UC-10; no automatic request or reservation follows.

Exception flows

- E1 at step 1: Unknown mandatory specifications exclude the asset until reviewed.
- E2 at step 2: Service failure or low-quality output is labelled and ordinary search remains usable.

Success postconditions A reviewed recommendation record exists; selection does not hold stock.

Failure postconditions No invalid recommendation overrides approval or quantity rules.

Rules BR-01, BR-04, BR-10, BR-11, BR-12. See section 2.6 for rule definitions.

Requirement links FR-11.1, FR-11.2, FR-11.3, FR-11.4, FR-11.4a, FR-11.4b, FR-12.1, FR-12.1a, FR-12.1b, FR-12.2, FR-12.2a, FR-12.3, FR-12.4, FR-12.4a, NFR-06, NFR-06a, NFR-07, NFR-09, NFR-13, NFR-13a, NFR-13b, NFR-13c, NFR-13d, NFR-17

UC 32 Review sustainable actions

Description and goal Compare AI suggestions for continued use or recovery of an asset.

Primary actor Sustainability officer

Supporting actors Asset custodian, External AI service

Preconditions The actor can access relevant condition, demand and evidence.

Trigger The actor selects Suggest next action.

Main flow

1. The system retrieves permitted asset, repair and benefit evidence.
2. The AI service proposes actions with reasons, uncertainty and alternatives.
3. The system identifies missing cost, mass or carbon evidence.
4. The actor accepts, changes or rejects the suggestion and may submit a proposal through UC-27.

Alternative flows

- A1 at step 4: Disposal is suggested; review available reuse/recovery alternatives and require ordinary authorization before execution.

Exception flows

- E1 at step 2: AI failure returns a manual proposal route.
- E2 at step 3: Missing evidence remains Unknown; no invented cost or factor is accepted.

Success postconditions The advice and human review outcome are recorded without executing an action.

Failure postconditions No stock, ownership or safety decision is made by AI alone.

Rules BR-04, BR-09, BR-10, BR-11, BR-12. See section 2.6 for rule definitions.

Requirement links FR-13.1, FR-13.2, FR-13.2a, FR-12.1, FR-12.1a, FR-12.1b, FR-12.2, FR-12.2a, FR-12.3, FR-12.4, FR-12.4a, NFR-06, NFR-06a, NFR-09, NFR-13, NFR-13a, NFR-13b, NFR-13c, NFR-13d

UC 33 Use the campus assistant

Description and goal Obtain permitted answers or a reviewed resource-demand draft.
Primary actor Campus user
Supporting actors External AI service
Preconditions The user is signed in and has scope for the requested records.
Trigger The user sends a supported question or drafting request.
Main flow
1. The system identifies the task and retrieves only permitted records or approved policy material.
2. The AI service returns a grounded answer or an editable demand draft.
3. The system displays references, uncertainty, draft fields and relevant authorized navigation links.
4. The user reviews the answer or explicitly confirms a draft that the system submits through UC-11 validation.

Alternative flows
- A1 at step 3: The user corrects or rejects a draft; no demand is created until explicit confirmation.

Exception flows
- E1 at step 1: No approved policy or supporting record exists; state insufficient information.
- E2 at step 2: Timeout or quota failure preserves manual search/forms; instructions to bypass approvals cause no protected action.

Success postconditions The user receives permitted information or a validated demand record.
Failure postconditions No unconfirmed request, invented policy or unauthorized disclosure occurs.
Rules BR-01, BR-06, BR-10, BR-11, BR-12. See section 2.6 for rule definitions.
Requirement links FR-14.1, FR-14.1a, FR-14.1b, FR-14.1c, FR-14.1d, FR-14.1e, FR-14.1f, FR-14.2, FR-14.2a, FR-14.2b, FR-14.3, FR-14.3a, FR-12.1, FR-12.1a, FR-12.1b, FR-12.2, FR-12.2a, FR-12.3, FR-12.4, FR-12.4a, NFR-06, NFR-06a, NFR-09, NFR-13, NFR-13a, NFR-13b, NFR-13c, NFR-13d, NFR-14, NFR-14a, NFR-14b

UC 34 Validate financial benefit

Description and goal Qualify an avoided purchase and its associated costs.
Primary actor Finance officer
Supporting actors Procurement officer
Preconditions Completed reuse and permitted purchasing-need evidence are available.
Trigger The finance officer opens financial evidence review.
Main flow
1. The system displays completed quantity and the proposed distinct purchasing need.
2. The officer reviews quantity, replacement price and attributed transport/repair costs.
3. The system identifies duplicate allocations or missing evidence.
4. The officer records a review decision and the system recalculates qualified gross and net values.

Alternative flows
- A1 at step 3: Price is known but repair cost is not; retain known gross value and Unknown net benefit.
- A2 at step 4: Correct earlier evidence by appending a reasoned revision and recomputing totals.

Exception flows
- E1 at step 3: Incomplete/disputed reuse or reused purchase evidence prevents a confirmed benefit.
- E2 at step 4: Saving fails; previous evidence and totals remain consistent.

Success postconditions Qualified amounts have a reviewer, evidence allocation and reproducible calculation.
Failure postconditions No double-counted saving or workflow-history edit occurs.
Rules BR-01, BR-02, BR-12, BR-13. See section 2.6 for rule definitions.
Requirement links FR-15.2, FR-15.2a, FR-15.3, FR-15.3a, FR-15.3b, FR-15.4, FR-15.4a, FR-15.4b, FR-15.5, FR-15.5a, FR-15.5b, NFR-15, NFR-15a

UC 35 Validate environmental evidence

Description and goal Review mass, disposal baseline, life extension and carbon-factor evidence.
Primary actor Sustainability officer
Supporting actors None
Preconditions The officer has scope for the completed event and environmental evidence.
Trigger The officer opens environmental evidence review.
Main flow
1. The system shows recorded quantity, mass, baseline and available factor sources.
2. The officer verifies units, boundaries, source dates and estimation basis.
3. The system checks duplicate disposal episodes and missing required inputs.
4. The officer confirms or corrects the evidence; the system records review history and recomputes available estimates.

Alternative flows
- A1 at step 3: No sourced carbon factor exists; retain Unknown carbon and display other supported indicators.

Exception flows
- E1 at step 2: An AI-generated unsourced factor or incompatible boundary is rejected.
- E2 at step 4: Saving fails; previous reviewed evidence remains intact.

Success postconditions Each estimate identifies its evidence, boundary and uncertainty.
Failure postconditions No fabricated factor or duplicate waste-diversion episode is counted.
Rules BR-01, BR-02, BR-12, BR-13. See section 2.6 for rule definitions.
Requirement links FR-15.3, FR-15.3a, FR-15.3b, FR-15.4, FR-15.4a, FR-15.4b, FR-15.5, FR-15.5a, FR-15.5b, NFR-15, NFR-15a

UC 36 View sustainability indicators

Description and goal Inspect traceable reuse, repair, financial and environmental outcomes.
Primary actor Report viewer
Supporting actors None
Preconditions The viewer has authorized dashboard scope.
Trigger The viewer opens the indicators dashboard.
Main flow
1. The viewer selects period and department filters.
2. The system selects qualified completed events and computes the section 3 formulas.
3. The system displays values, units, coverage, estimate status and Unknown inputs.
4. The viewer follows an indicator to its permitted contributing records.

Alternative flows
- A1 at step 3: No qualifying events exist; show zero activity and distinguish this from unknown evidence.

Exception flows
- E1 at step 1: A restricted department filter is denied.
- E2 at step 2: Calculation or retrieval fails; report failure without inventing totals.

Success postconditions Displayed indicators reconcile with qualified records.
Failure postconditions Viewing does not alter source events or benefit evidence.
Rules BR-01, BR-12, BR-13. See section 2.6 for rule definitions.
Requirement links FR-15.1, FR-15.1a, FR-15.1b, FR-15.1c, FR-15.1d, FR-15.1e, FR-15.1f, FR-15.1g, FR-15.3, FR-15.3a, FR-15.3b, FR-15.4, FR-15.4a, FR-15.4b, NFR-15, NFR-15a

UC 37 Generate an operational report

Description and goal Obtain conventional reports for permitted records and periods.
Primary actor Report viewer
Supporting actors None
Preconditions The viewer has reporting permission.
Trigger The viewer selects a report type.
Main flow
1. The viewer selects date and department filters.
2. The system validates scope and retrieves authoritative records.
3. The system calculates rows and totals for the selected report.
4. The system displays the result with period, units and qualification rules.

Alternative flows
- A1 at step 3: No records match; return an explicitly empty report.

Exception flows
- E1 at step 2: Unauthorized scope is denied without aggregate disclosure.
- E2 at step 3: Retrieval or calculation fails; offer retry without presenting a partial total as complete.

Success postconditions Report rows and totals match saved records.
Failure postconditions No source workflow or evidence record changes.
Rules BR-01, BR-11, BR-12, BR-13. See section 2.6 for rule definitions.
Requirement links **FR-17.1, FR-17.3, FR-17.3a, FR-17.3b, NFR-15, NFR-15a, NFR-12, NFR-12a**

UC 38 Review an AI report summary

Description and goal Produce a human-reviewed narrative of verified report figures.
Primary actor Report viewer
Supporting actors External AI service
Preconditions A conventional report from UC-37 is available within scope.
Trigger The viewer selects Generate summary.
Main flow
1. The system sends permitted verified figures and estimate labels to the AI service.
2. The service returns a summary with report references and model metadata.
3. The viewer checks each numeric claim, qualification and interpretation.
4. The viewer accepts or corrects the wording; the system saves the summary with review and model records.

Alternative flows
- A1 at step 3: Unsupported wording appears; revise or reject it before saving.

Exception flows
- E1 at step 2: Generation fails or times out; keep the conventional report available.
- E2 at step 4: Saving fails; do not label the draft as a saved reviewed summary.

Success postconditions A reviewed summary is traceable to the source report and model record.
Failure postconditions No fabricated number becomes an accepted report fact.
Rules BR-01, BR-10, BR-11, BR-12, BR-13. See section 2.6 for rule definitions.
Requirement links **FR-17.2, FR-17.3, FR-17.3a, FR-17.3b, FR-12.1, FR-12.1a, FR-12.1b, FR-12.2, FR-12.2a, FR-12.3, FR-12.4, FR-12.4a, NFR-06, NFR-06a, NFR-09, NFR-15, NFR-15a**

UC 39 Read notifications

Description and goal Inspect event notifications and mark them read.
Primary actor Campus user
Supporting actors None
Preconditions The user is signed in; notifications may exist for that recipient.
Trigger The user opens notifications.
Main flow
1. The system retrieves notifications for the current recipient.
2. The user selects a notification.
3. The system rechecks permission for its linked record and displays permitted details.
4. The user marks it read and the system saves the read state.

Alternative flows
- A1 at step 1: No notifications exist; show an empty inbox without error.

Exception flows
- E1 at step 3: Access to the linked record has been revoked; withhold its details.
- E2 at step 4: Saving read state fails; retain unread status. Delivery failures remain retryable and never reverse the original business event.

Success postconditions The recipient sees permitted notifications; read status is saved when confirmed.
Failure postconditions No other recipient data or business state is changed.
Rules BR-01, BR-14. See section 2.6 for rule definitions.
Requirement links **FR-16.1, FR-16.2, FR-16.2a, FR-16.2b, FR-16.2c**

UC 40 Search audit records

Description and goal Review important changes and denied privileged actions.
Primary actor Asset administrator
Supporting actors None
Preconditions The administrator has authorized audit scope.
Trigger The administrator opens audit search.
Main flow
1. The administrator enters actor, target, date or action filters.
2. The system applies audit permissions and retrieves matching events.
3. The system displays actor, time, outcome and relevant before/after values.
4. The administrator follows permitted record references without modifying the events.

Alternative flows
- A1 at step 3: No matching events exist; revise filters and return to step 1.

Exception flows
- E1 at step 2: Unauthorized audit scope is denied.
- E2 at step 4: An edit/delete attempt is refused; retrieval failure produces a retryable error.

Success postconditions The audit trail supports inspection with original events intact.
Failure postconditions No event is erased or unauthorized history disclosed.
Rules BR-01, BR-02, BR-15. See section 2.6 for rule definitions.
Requirement links **FR-16.3, FR-16.4, FR-16.4a**

UC 41 Inspect asset history

Description and goal Trace an asset through acquisition, use, service, transfer and disposition.
Primary actor Campus user
Supporting actors None
Preconditions The user can access the asset and the requested history fields.
Trigger The user opens Asset history.
Main flow
1. The system retrieves chronological events and parent/child lot links.
2. The user filters event types or dates.
3. The system distinguishes current state, historical events and later corrections.
4. The user follows a permitted maintenance, allocation, transfer or disposition reference.

Alternative flows
- A1 at step 1: A record is archived or an actor deactivated; preserve and display the historical identifier.

Exception flows
- E1 at step 4: Restricted financial or administrator detail is withheld.
- E2 at step 1: Retrieval failure is reported without inventing missing history.

Success postconditions The asset and quantity movements are traceable within the user scope.
Failure postconditions History and stock remain unchanged.
Rules BR-01, BR-02, BR-03. See section 2.6 for rule definitions.
Requirement links **FR-08.1, FR-02.4, FR-02.4a, FR-02.4b, FR-16.4, FR-16.4a**

8 Priorities traceability and verification

8.1 Priority rationale and extensions

High/P0 identifies the required foundation, and Medium/P1 identifies required intelligence and reporting. Low/Desirable extensions and Future enhancements are listed separately. These priorities express build order and risk while retaining the full brief's required classic and AI scope. Source review and implementation may refine the order without deleting a mandatory feature.

Priority	Meaning	Examples
High / P0	Essential foundation, delivered first	Identity, data, publication, demand, approval/reservation, transfers, maintenance and audit
Medium / P1	Essential completion after the foundation	Classification, matching, explanations, suggestions, assistant, indicators and summaries
Low / Desirable	Useful extension after essential acceptance	Saved search, email channel, CSV export, bulk import, personal dashboard, multi-item convenience
Future	Outside this prototype scope	Live campus integrations, native mobile app, production-scale deployment

ID	Extension	Priority	Acceptance or boundary
EXT-01	Saved search filters	Desirable	Store and reload a named filter set for the signed-in user.
EXT-02	Email notifications	Desirable	Send the existing event notifications through an additional email channel without changing workflow outcomes.
EXT-03	CSV report export	Desirable	Export authorized report rows with the same filters, units and permission scope as the on-screen report.
EXT-04	Bulk asset import	Desirable	Validate a preview of imported asset rows before confirming creation and identify rejected rows.
EXT-05	Personal dashboard layout	Desirable	Let a user rearrange permitted dashboard panels without changing their calculations.
EXT-06	Multi-item request convenience	Desirable	Group several independently validated item requests under one user-facing reference while retaining separate quantities and approvals.
FUT-01	Live campus integrations	Future	Integrate university identity, procurement and asset systems only after institutional approval.
FUT-02	Native mobile application	Future	Provide a separate mobile application after the browser prototype is accepted.
FUT-03	Production scale deployment	Future	Add institution-wide operations, service levels and controls after a separate readiness assessment.

8.2 Requirement coverage

Parent FR	Detailed IDs	Related use cases
FR-01	FR-01.1 to FR-01.4 (including letter-suffixed requirements)	UC-01, UC-02, UC-03, UC-04, UC-05
FR-02	FR-02.1 to FR-02.5 (including letter-suffixed requirements)	UC-06, UC-07, UC-22, UC-41

Parent FR	Detailed IDs	Related use cases
FR-03	FR-03.1 to FR-03.3 (including letter-suffixed requirements)	UC-08, UC-09
FR-04	FR-04.1 to FR-04.4 (including letter-suffixed requirements)	UC-11, UC-12, UC-13, UC-14, UC-22
FR-05	FR-05.1 to FR-05.3 (including letter-suffixed requirements)	UC-10
FR-06	FR-06.1 to FR-06.5 (including letter-suffixed requirements)	UC-13, UC-14, UC-15, UC-16, UC-18, UC-19, UC-20
FR-07	FR-07.1 to FR-07.6 (including letter-suffixed requirements)	UC-14, UC-17, UC-18, UC-19, UC-20, UC-21, UC-22, UC-23
FR-08	FR-08.1 to FR-08.4 (including letter-suffixed requirements)	UC-06, UC-23, UC-25, UC-27, UC-28, UC-29, UC-41
FR-09	FR-09.1 to FR-09.4 (including letter-suffixed requirements)	UC-24, UC-25, UC-26, UC-27
FR-10	FR-10.1 to FR-10.3 (including letter-suffixed requirements)	UC-30
FR-11	FR-11.1 to FR-11.4 (including letter-suffixed requirements)	UC-31
FR-12	FR-12.1 to FR-12.4 (including letter-suffixed requirements)	UC-30, UC-31, UC-32, UC-33, UC-38
FR-13	FR-13.1 to FR-13.2 (including letter-suffixed requirements)	UC-32
FR-14	FR-14.1 to FR-14.3 (including letter-suffixed requirements)	UC-33
FR-15	FR-15.1 to FR-15.5 (including letter-suffixed requirements)	UC-34, UC-35, UC-36
FR-16	FR-16.1 to FR-16.4 (including letter-suffixed requirements)	UC-03, UC-04, UC-05, UC-07, UC-12, UC-15, UC-16, UC-17, UC-21, UC-24, UC-39, UC-40, UC-41
FR-17	FR-17.1 to FR-17.3 (including letter-suffixed requirements)	UC-37, UC-38

Appendix A contains one row for each of the 189 detailed requirements, linking its source, priority, use cases, rules, verification prerequisites and acceptance-check identifier. The full acceptance procedures remain beside the requirements in sections 4 and 5. Execution status is Planned / not executed for every check. The report does not depend on separate traceability or acceptance-check files.

8.3 Use case identification

Use cases UC-01 through UC-41 are defined in section 7. Appendix B lists their primary and supporting actors; Appendix A links them to requirements and acceptance checks. Use these identifiers consistently in design, implementation and test records.

8.4 Dependency and verification order

Build identity and data before publication/search and demand. Add allocation requests and destination/source approval before transfers and receipt. Maintenance and disposition depend on trustworthy asset state. AI uses those permitted records and manual workflows. Indicators and summaries depend on completed events and reviewed evidence. Verification prerequisites for each detailed requirement are recorded in Appendix A.

Evaluation controls: two team reviewers will label matching relevance independently before seeing model output. Hold ten of the forty requests and fifty labelled assets out of tuning. Record disagreements, model/configuration, IDs, exclusions and all failures. Apply NFR-17 to five paired requests differing only in department names, with identical permissions, locations and scoring inputs. Report limited sample sizes without extrapolating campus-wide accuracy or fairness. Reviewer judgments and test outcomes remain to be collected during evaluation.

- Check submission leaves eight monitors Free; destination acceptance still leaves eight; final source approval of five produces three Free and five Held.
- Attempt two final approvals concurrently against insufficient stock; only the qualifying transaction may reserve, with no partial decision or hold.
- Exercise expiry at the exact 48-hour boundary, cancellation before/after approval and blocked post-dispatch cancellation.
- Exercise allowed and denied role paths, including self-approval, distinct approvers, direct URLs and AI-assisted actions.
- Exercise dispatch, receipt, discrepancies, confirmed return, maintenance clearance and authorized actual disposition.
- Recalculate each sustainability indicator with duplicate, missing and negative-value examples; preserve Unknown values.
- Compare matching and classification against fixed held-out labels; record failures, latency and manual fallback.
- Use actual tester, date, result and evidence when execution occurs during later phases; record defects and retests.

9 Evidence decisions and handover

9.1 Open decisions

The following operating defaults make the academic specification concrete. They are not verified university policy. Resolve material decisions through actual team/instructor review and update requirements, checks and history if a choice changes.

ID and decision	Selected rule or unresolved issue	Owner and target
D-01 Approval roles and order	Destination acceptance precedes source-custodian release; final release creates the hold. Distinct approvers cannot be the requester. Status: Selected prototype rule; campus authority still to confirm.	Ali; 12 October 2026
D-02 Pending request expiry	Retain the proposed 48-hour response window for non-binding pending allocation requests; approved reservations do not expire automatically. Status: Prototype rule aligned with S11; campus confirmation remains pending.	Ali; 12 October 2026
D-03 Evidence and sample data	200 assets, 40 requests, 20 transfer and 20 maintenance examples; fictional departments and identities. Status: Instructor acceptance to confirm.	Ahmed; 13 October 2026
D-04 AI provider and timeout	Provider/model selected after trial; interactive timeout 15 seconds; provider version may be unavailable. Status: Provider and threshold need trial.	Ahmed; 14 October 2026
D-05 File and interface limits	Five attachments per asset; JPEG/PNG/PDF; 5 MiB each; English browser prototype. Status: New explicit prototype limits.	Suhail; 13 October 2026
D-06 Retention and real data	Delete hosted fictional personal data and backups within 90 days after grading closes unless a documented course retention obligation applies. NFR-13c defines the testable rule. Real-data use remains disabled pending institutional PDPL and provider/cross-border review under NFR-13d.	Ali; 16 October 2026
D-07 Conflicting deadlines	S9 slide 2 states labs in the week of 5 October 2026; S1 p 10 states the week beginning 18 October. Neither is assumed to supersede the other. Obtain the actual deadline and record any revised submission arrangement. Status: Unresolved; do not treat 18 October as a confirmed deadline.	Suhail; 12 October 2026
D-08 Demonstration interviews	Completed: two role-based demonstration interviews on 12 October 2026, following the instructor-approved approach reported by the team. Findings and requirement links are recorded in section 9.2. Actual campus policy validation is separate.	Completed 12 October 2026; team review of findings remains pending.
D-09 Repository and review	Use the supplied shared repository; verify access, existing conventions, instructor/lab-engineer collaboration and actual contribution history. Status: Repository access verification pending.	Ahmed with team; 12 October 2026
D-10 Safety and disposition documents	Use sample clearance and handler evidence; actual hazardous-equipment and disposal policy must be confirmed before real use. Status: Prototype demonstration only.	Ali; 14 October 2026

9.2 Feedback and stakeholder evidence

Feedback topic	Current response	Status
Dates missing	Dated report and follow-up tasks; source conflict recorded explicitly	Documented; actual submission deadline unresolved
Stakeholder interviews	Two completed demonstration interviews and a findings-to-requirement register in section 9.2; the method follows the instructor instruction reported by the team.	Demonstration activity completed 12 October 2026; no claim of external human interviews.
Three systems	Snipe-IT, Warp It and Rheaply comparison with official references	Desk research documented
Numerical benefit	AED 2300 illustrative net benefit with qualification rules	Hypothetical estimate, not observed savings
Nearly all High priority	High/P0 versus Medium/P1 plus Low/Desirable and Future	Defined with rationale; team review pending
Out-of-scope too short	Explicit exclusion list in section 2.2	Documented

Two written demonstration interviews were completed on 12 October 2026: Interview A represents an asset custodian, and Interview B represents a departmental requester. The participants are role-play personas for the course demonstration; the answers below are stakeholder perspectives, not verified university practices or policies. The instructor-approved use of this approach was reported by the team. No personal identities or employee affiliations are claimed. [S12-S13]

9.2.1 Interview A with an asset custodian

Question 1 What would you want this system to improve?

Answer I need to know which assets are available, who is responsible for them and where they are. When another department asks for an item, I want the request and handover recorded in one place so I can follow what happened without reconstructing it from separate messages.

Question 2 What information would you check before listing an asset?

Answer I would check its name, specifications, quantity, condition, location and assigned custodian. A photograph or supporting document would help me judge suitability. If the condition is unknown or an inspection is still needed, I would keep it unavailable until someone authorized records clearance.

Question 3 When should requested stock become reserved?

Answer I would not reserve it just because someone submits a request. The destination should accept the need first, then I should approve release. At that final step, the system must check that the units are still free and reserve them together with the approval. If two requests compete for the same stock, both must not succeed.

Question 4 Who should be allowed to approve and receive an item?

Answer The requester should not approve their own request. I would keep destination acceptance and source release with different people, then ask the receiving custodian to confirm quantity and condition when the item arrives. Technical administration alone should not give someone business approval authority.

Question 5 How would you handle expiry and cancellation?

Answer A visible response deadline would help me manage the queue. For the demonstration, 48 hours is a workable assumption for a pending request. Once I approve and reserve stock, I would not want that hold to disappear because the original response deadline passes. A cancellation before dispatch should release the hold once; after dispatch it needs a return or dispute process.

Question 6 What should happen if the delivery is damaged or incomplete?

Answer I would want the receiver to report the difference with supporting evidence. The transfer should remain disputed until an authorized person records accepted receipt or return. The system must account for every unit and should not count a disputed transfer as a successful reuse benefit.

Question 7 How should maintenance and disposal be controlled?

Answer A safety concern should block allocation and dispatch. Maintenance staff should record their findings, work and final condition. If repair is unsuitable, I can propose another action, but a different authorized person should approve consequential disposition. The actual handler, date and quantity should be recorded when the action is carried out.

Question 8 What would make AI recommendations useful to you?

Answer I would use suggestions if I can see why an asset or action was recommended. Show the specifications, condition, location, costs and missing evidence. Let me correct or reject a suggestion. AI should never decide that equipment is safe or authorize release or disposal for me.

Question 9 What would you need when AI is unavailable?

Answer I still need to register assets, search, review requests, approve transfers and read normal reports. Tell me that the AI service is unavailable and keep my valid work so I can continue manually.

Question 10 Which records matter most for accountability?

Answer Keep the sequence of approvals, dispatch, receipt, maintenance and disposition, including who acted and when. Corrections should remain visible. Notifications should take each person to the record they are allowed to see, without exposing unrelated departmental information.

9.2.2 Interview B with a departmental requester

Question 1 How would you begin looking for a resource?

Answer I would search existing stock before asking for a new purchase. I want to filter by category, condition, location and available quantity. I also want to describe what I need in ordinary language when I do not know the exact asset name.

Question 2 What should a resource request contain?

Answer I would explain the purpose, mandatory specifications, quantity, minimum condition, delivery location, urgency and required date. Preferences should be separate from requirements. A cheaper computer is not a suitable substitute if it cannot meet the required memory or software needs.

Question 3 How should the system explain a recommendation?

Answer Show me the compatible specifications, quantity, condition, location and relevant costs. Explain any missing information and let me compare alternatives. A high score should not look like a guarantee that the item is suitable.

Question 4 How should equivalent requests from different departments be treated?

Answer Changing only a department name should not change the result. Differences in real needs, urgency, location or permissions can matter, but the reason should be recorded. I would want unexplained differences investigated before trusting the ranking.

Question 5 What do you need to know while waiting for a request?

Answer Show whether the destination has accepted it, whether the source has approved it, what is reserved and whether it has been dispatched. A reserved item is not the same as an item I have received. If my request is rejected or expires, show the reason and what I can do next.

Question 6 Would partial fulfilment be useful?

Answer Yes. If I need ten chairs and receive four, I want the request to show four received and six still needed. It should separately show any remaining commitments so I do not accidentally request more than I need.

Question 7 How should the assistant help you?

Answer It should answer permitted questions, prepare a request for me to check, and guide me to the correct form or report. I want to edit the fields and confirm before anything is submitted. If I ask about a policy that is not available, it should say that rather than invent an answer.

Question 8 What sustainability information would you find convincing?

Answer I would want to see the underlying evidence. Savings should use a real purchasing need, replacement-price evidence and the relevant transfer and repair costs. Reuse rate should say what was offered and what was received by the cutoff. Missing weights or emission factors should stay unknown, and repeated transfers should not create duplicate savings.

Question 9 What makes the system practical to use?

Answer Give me clear field errors, visible status and understandable next steps. I should be able to complete the core forms using a keyboard. Search and requests should still work when AI is unavailable.

Question 10 Which features should come first?

Answer Accurate stock, search, requests, permissions, approvals and reliable handover should come first. AI matching and useful sustainability reports matter too, but I would rather have a dependable core workflow before extras such as saved searches or email notifications.

9.2.3 Interview findings and requirement decisions

The following findings connect the demonstration interviews to the requirements in this specification. Each decision identifies the applicable requirement, rule or verification criterion. Demonstration agreement does not replace team approval or real-campus validation.

Finding	Question source	Need	Requirement decision
INT-01	A2; B2	Complete metadata and mandatory suitability	FR-02.1a, FR-04.1, FR-11.1; BR-04.
INT-02	A3-A4	Approval-time reservation and separate decision-makers	FR-06.2, FR-06.2a, FR-07.1; BR-08; NFR-05.
INT-03	A5; B5-B6	Clear expiry, cancellation and fulfilment status	BR-05, BR-06; FR-04.3; UC-14, UC-16, UC-19.
INT-04	A6-A7	Safety, disputes and evidenced disposition	FR-07, FR-08, FR-09; NFR-16; UC-23.
INT-05	A8; B3	Explain AI output and allow human correction	FR-12.1 to FR-12.4; UC-30 to UC-33.
INT-06	A9; B9	Manual continuity and usable core workflows	FR-05.3; NFR-02, NFR-06a, NFR-11.
INT-07	B4	Equivalent departmental treatment	NFR-17 and AC-NFR-17; controlled paired comparison in section 8.4.
INT-08	B7	Assistant navigation and confirmation	FR-14.1f, AC-FR-14.1f and UC-33 step 3; FR-14.2–FR-14.2b.
INT-09	B8	Explicit reuse rate and evidence-based benefits	FR-15.1a and AC-FR-15.1a; section 3.4 formulas; FR-15.2 to FR-15.5 including letter-suffixed requirements.
INT-10	A10; B10	Auditability and core-first implementation	FR-16; section 2.8 notification matrix; section 8.1 priorities.

Interview conclusion: both demonstration roles support the existing controlled workflow. The interview findings address fairness checks, assistant navigation and explicit reuse-rate verification. The interview activity is complete; team review of these findings, campus authority confirmation and application testing remain separate tasks.

9.3 Conflicting dates and dated follow up

S9 slide 2 lists Phase 3 in labs during the week of 5 October 2026. S1 page 10 lists the week beginning 18 October 2026. As of 12 October 2026, one listed date has already passed. Neither document proves that an extension exists. Confirm the actual deadline and any necessary submission arrangement with the instructor; the internal dates below are planning targets only.

Date	Planned action	Owner and output
11 October 2026	Prepare the requirements report using the teaching materials	Prepared report; human review pending
12 October 2026	Confirm deadline and review selected approval/expiry model	Suhail/Ali: D-01, D-02 and D-07 decisions
12 October 2026	Complete two demonstration interviews and document their findings	Completed: Interview A, Interview B, findings register and integrated appendices; human review pending.
13 October 2026	Review scope, roles, required fields and diagrams	Team: actual issue/review record
14 October 2026	Review completed demonstration interview findings and AI feasibility evidence	Team review pending; interviews completed 12 October; D-04 and D-10 remain open.
15 October 2026	Review the interview findings, requirements and acceptance criteria	Findings INT-01 to INT-10 are documented; team approval and any further corrections remain pending.
16 October 2026	Complete cross-reference and submission review	Team: reviewed report and D-06
Instructor-confirmed date	Submit through the actual course channel	Team: submission receipt; do not assume 18 October

9.4 Team review and AI assistance

Proposed review responsibilities are: Suhail reviews scope, requirements and instructor format; Ali reviews records, authority, stock and state transitions; Ahmed reviews AI, evaluation and benefit calculations. A real review record must contain reviewer, date, affected IDs, issue, decision and resolution evidence. These assignments do not claim that reviews have occurred.

ChatGPT assisted with source analysis, official-system comparison, drafting, consistency checks, diagram construction and report preparation on 11–12 October 2026. This report records two completed role-play stakeholder interviews for the course demonstration. Team members remain responsible for understanding, reviewing and explaining the submission and validating later implementation. Demonstration interview answers do not establish actual campus policy, human approval or passed application tests.

9.5 Repository and submission handover

Repository work is tracked separately from this requirements report. Target: https://github.com/A7mdmk/Asset-Management-Project-. This report does not claim a repository inspection, push, collaborator invitation or completed review. When resumed, preserve earlier submissions and existing history, use each contributor’s verified identity and commit genuine changes. The course brief requires meaningful ongoing contributions; the report is not evidence that those contributions have occurred. [S1, S4]

- Confirm the controlling deadline and submission format.

- Review the two completed demonstration stakeholder interviews and their findings register in section 9.2; retain their role-play designation when presenting the course demonstration.
- Review the selected approval-on-reservation timing and all open decisions.
- Keep this report, the embedded diagram, appendices and actual review evidence consistent.
- Follow the course AI-use rules and be prepared to explain the requirements and use cases.
- Resume repository access and contribution checks separately; submit through the required course channel.

10 References and glossary

[S1] Khalifa University COSC 336. Project September 2026. Supplied full project brief; Phase 3 on page 8, Git requirements on pages 9-10, schedule on page 10 and SRS guidance on pages 11-19.

[S2] KU-ResourceSavers. Phase 1. Dated 14 September 2026. Earlier team report; initial FR-01 to FR-17, NFR-01 to NFR-10, success criteria and team walkthrough. Source copy is not included with this report; the citation has not been independently rechecked.

[S3] KU-ResourceSavers. Phase 2 updated. Earlier feasibility document, cover dated 15 September 2026 with later AI-log entries. Sections 2-4, 6-7, 11-12 and assumptions used; selected tools and policy remain proposals. Source copy is not included with this report; the citation has not been independently rechecked.

[S4] Git GitHub Copilot Onboarding. Earlier course onboarding guide, 3 September 2026. Repository setup, meaningful commits, branches, review and AI-use responsibilities. Source copy is not included with this report; the citation has not been independently rechecked.

[S5] Instructor feedback on Phase 1, relayed by the team on 11 October 2026. Dates, two interviews, three systems, benefit estimate, prioritization and scope exclusions.

[S6] Snipe-IT. Product Features. https://snipeitapp.com/product . Accessed 11 October 2026.

[S7] Warp It. The Reuse Network. https://www.warp-it.co.uk/ . Accessed 11 October 2026.

[S8] Rheaply. Workplace Resources. https://rheaply.com/solutions/workplace-resources/ . Accessed 11 October 2026.

[S9] COSC336. Phase 3 Slides falll26, Fall 2026. Supplied 26-slide deck; slide 2 schedule, slides 13-16 use-case fields, slide 26 report structure and SRS references.

[S10] Functional and Non-Functional Requirements. Requirements Exercises worksheet. Classification and testability examples; numerical targets are explicitly classroom examples. Source copy is not included with this report; the citation has not been independently rechecked.

[S11] Use Case Writing Exercises. Teaching worksheet. Search, submit and review examples with separate alternatives/exceptions; submission reserves no stock, and successful approval reserves quantity. Scenarios are explicitly classroom assumptions. Source copy is not included with this report; the citation has not been independently rechecked.

[S12] Stakeholder demonstration interview A. Asset-custodian role. Written role-play completed 12 October 2026; full questions and answers in section 9.2.1; requirement findings in section 9.2.3.

[S13] Stakeholder demonstration interview B. Departmental-requester role. Written role-play completed 12 October 2026; full questions and answers in section 9.2.2; requirement findings in section 9.2.3.

Term	Meaning
Demand	A departmental need with requested and received quantity; it may be fulfilled from several assets.
Allocation request	A non-binding request for a specified asset quantity against demand; pending submission holds no stock.
Reservation	An approved quantity hold created only when final source approval succeeds.
Transfer	The recorded movement from source dispatch to accepted receipt or resolved return.

Term	Meaning
Custody	Recorded departmental responsibility and physical location; legal university ownership does not change in an internal transfer.
Atomic and idempotent	A change applies completely or not at all; repeating an accepted event does not apply it twice.
Alternative and exception	An alternative is a specified branch of normal business interaction; an exception describes failure or a denied operation and its recovery.
Unknown	Missing evidence; distinct from a measured value of zero.
Compatibility and confidence	A ranking score orders candidates; it is not automatically a calibrated probability.
Traceability	Links between evidence, requirements, user goals, rules and verification criteria.

[S14] W3C. Web Content Accessibility Guidelines (WCAG) 2.1, W3C Recommendation, 6 May 2025; Level AA conformance includes all Level A and AA criteria. https://www.w3.org/TR/WCAG21/ . Accessed 14 October 2026.

[S15] UAE. Federal Decree-Law No. 45 of 2021 Concerning the Protection of Personal Data (UAE PDPL). Official legislation: https://uaelegislation.gov.ae/en/legislations/1972 ; official overview: https://u.ae/en/about-the-uae/digital-uae/data/data-protection-laws . Accessed 14 October 2026. Institutional applicability and any exclusions require review before real-data use; the prototype controls do not certify legal compliance.

[S16] U.S. Environmental Protection Agency. WARM, Electronics, December 2023, Exhibit 1-5, p. 1-4. Desktop CPU source-reduction factor: -20.86 MTCO2e/short ton. https://www.epa.gov/system/files/documents/2023-12/warm_electronics_v16_dec.pdf . Accessed 14 October 2026.

10.1 Source availability and use

Sources	Availability and permitted use
S1 and S9	Supplied project brief and Phase 3 slide deck. They establish the current requirements scope and submission structure. Their schedules disagree; D-07 remains unresolved.
S2-S4 and S10-S11	Supporting reports, Git guidance and teaching worksheets. Copies are not included with this report, and their contents have not been independently rechecked.
S5	Instructor feedback relayed by the team. The team also reports that role-play stakeholder interviews are permitted for this demonstration; no written instructor confirmation is reproduced.
S6-S8	Public comparison URLs with a recorded access date of 11 October 2026. The comparison is desk research, not evidence of a hands-on product trial or measured campus need. The access records have not been independently rechecked.
S12-S13	Complete demonstration interview transcripts and findings are included in section 9.2. They support prototype requirement reasoning and do not verify actual university practices.
Phase 2 teaching slides	The attached feasibility-study slide deck is teaching guidance, not the earlier team feasibility report referenced as S3. These sources must not be treated as interchangeable.
S14-S16	Official WCAG 2.1, UAE PDPL references and EPA WARM factor sources are named in section 10; acceptance controls and the carbon example are stated in full in this report.

Requirements needed to understand the prototype are stated within this document even when an earlier source is unavailable. Any material discrepancy found when source copies are checked must be recorded against the affected requirement or decision. No unavailable source is silently treated as confirmed.

Appendix A Requirement traceability and verification register

Each row identifies one requirement and its acceptance check. Full requirement and check wording remains in sections 4 and 5. Source IDs are defined in section 10; INT IDs refer to the demonstration findings in section 9.2.3. Earlier-source references retain the limitations in section 10.1.

The check for each row is AC- followed by its requirement ID, for example AC-FR-01.1 or AC-NFR-17. Every check has status Planned / not executed. On execution, record tester, date, environment/model, fixture IDs, actual result, evidence location, defect and retest outcome against that check ID. Blank execution evidence is not a pass.

P0 denotes a foundation requirement and P1 denotes required intelligence or reporting; both are essential. Quality requirements apply wherever relevant. Prerequisites below are verification inputs, not an imposed software architecture. Use-case links are taken from section 7; Cross-cutting means the requirement is verified across the stated workflow rather than a separate user goal.

A.1 Functional requirements

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-01.1 P0	S1, S2, S3	UC-03, UC-04	BR-01 Role and department fixtures
FR-01.1a P0	S1, S2, S3	UC-03, UC-04	BR-01 Role and department fixtures
FR-01.1b P0	S1, S2, S3	UC-03, UC-04	BR-01 Role and department fixtures
FR-01.1c P0	S1, S2, S3	UC-03, UC-04	BR-01 Role and department fixtures
FR-01.2 P0	S1, S2, S3	UC-01, UC-02	Feature and state rules Role and department fixtures
FR-01.2a P0	S1, S2, S3	UC-01, UC-02	Feature and state rules Role and department fixtures
FR-01.3 P0	S1, S2, S3	UC-01, UC-04	BR-01 Role and department fixtures
FR-01.4 P0	S1, S2, S3	UC-03, UC-05	BR-02 Role and department fixtures
FR-01.4a P0	S1, S2, S3	UC-03, UC-05	BR-02 Role and department fixtures
FR-02.1 P0	S1, S2, S3; INT-01	UC-06	Feature and state rules FR-01; asset field fixtures
FR-02.1a P0	S1, S2, S3; INT-01	UC-06	Feature and state rules FR-01; asset field fixtures
FR-02.2 P0	S1, S2, S3; INT-01	UC-07	BR-02, BR-03 FR-01; asset field fixtures
FR-02.2a P0	S1, S2, S3; INT-01	UC-07	BR-02, BR-03 FR-01; asset field fixtures

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-02.2b P0	S1, S2, S3; INT-01	UC-07	BR-02, BR-03 FR-01; asset field fixtures
FR-02.3 P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-15 FR-01; asset field fixtures
FR-02.3a P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-15 FR-01; asset field fixtures
FR-02.3b P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-15 FR-01; asset field fixtures
FR-02.3c P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-15 FR-01; asset field fixtures
FR-02.3d P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-15 FR-01; asset field fixtures
FR-02.4 P0	S1, S2, S3; INT-01	UC-06, UC-22, UC-41	BR-03, BR-07 FR-01; asset field fixtures
FR-02.4a P0	S1, S2, S3; INT-01	UC-06, UC-22, UC-41	BR-03, BR-07 FR-01; asset field fixtures
FR-02.4b P0	S1, S2, S3; INT-01	UC-06, UC-22, UC-41	BR-03, BR-07 FR-01; asset field fixtures
FR-02.5 P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-01 FR-01; asset field fixtures
FR-02.5a P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-01 FR-01; asset field fixtures
FR-02.5b P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-01 FR-01; asset field fixtures
FR-02.5c P0	S1, S2, S3; INT-01	UC-06, UC-07	BR-01 FR-01; asset field fixtures
FR-03.1 P0	S1, S2, S3; INT-01	UC-08	BR-03, BR-04 FR-01, FR-02; BR-04
FR-03.2 P0	S1, S2, S3; INT-01	UC-08, UC-09	BR-04 FR-01, FR-02; BR-04
FR-03.2a P0	S1, S2, S3; INT-01	UC-08, UC-09	BR-04 FR-01, FR-02; BR-04
FR-03.3 P0	S1, S2, S3; INT-01	UC-09	BR-04, BR-05 FR-01, FR-02; BR-04
FR-03.3a P0	S1, S2, S3; INT-01	UC-09	BR-04, BR-05 FR-01, FR-02; BR-04
FR-03.3b P0	S1, S2, S3; INT-01	UC-09	BR-04, BR-05 FR-01, FR-02; BR-04

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-04.1 P0	S1, S2, S3; INT-01, INT-03	UC-11	BR-15 FR-01, FR-02; demand fixtures
FR-04.2 P0	S1, S2, S3; INT-01, INT-03	UC-12	BR-06 FR-01, FR-02; demand fixtures
FR-04.2a P0	S1, S2, S3; INT-01, INT-03	UC-12	BR-06 FR-01, FR-02; demand fixtures
FR-04.3 P0	S1, S2, S3; INT-01, INT-03	UC-14, UC-22	BR-06, BR-07 FR-01, FR-02; demand fixtures
FR-04.3a P0	S1, S2, S3; INT-01, INT-03	UC-14, UC-22	BR-06, BR-07 FR-01, FR-02; demand fixtures
FR-04.4 P0	S1, S2, S3; INT-01, INT-03	UC-13	BR-05, BR-06 FR-01, FR-02; demand fixtures
FR-04.4a P0	S1, S2, S3; INT-01, INT-03	UC-13	BR-05, BR-06 FR-01, FR-02; demand fixtures
FR-04.4b P0	S1, S2, S3; INT-01, INT-03	UC-13	BR-05, BR-06 FR-01, FR-02; demand fixtures
FR-04.4c P0	S1, S2, S3; INT-01, INT-03	UC-13	BR-05, BR-06 FR-01, FR-02; demand fixtures
FR-05.1 P0	S1, S2, S3; INT-06	UC-10	BR-01 FR-01, FR-02, FR-03
FR-05.1a P0	S1, S2, S3; INT-06	UC-10	BR-01 FR-01, FR-02, FR-03
FR-05.2 P0	S1, S2, S3; INT-06	UC-10	Feature and state rules FR-01, FR-02, FR-03
FR-05.2a P0	S1, S2, S3; INT-06	UC-10	Feature and state rules FR-01, FR-02, FR-03
FR-05.2b P0	S1, S2, S3; INT-06	UC-10	Feature and state rules FR-01, FR-02, FR-03
FR-05.3 P0	S1, S2, S3; INT-06	UC-10	BR-11 FR-01, FR-02, FR-03
FR-05.3a P0	S1, S2, S3; INT-06	UC-10	BR-11 FR-01, FR-02, FR-03
FR-06.1 P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-15	BR-03, BR-06 FR-01 to FR-04; approval fixtures
FR-06.1a P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-15	BR-03, BR-06 FR-01 to FR-04; approval fixtures
FR-06.2 P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-18	BR-03 FR-01 to FR-04; approval fixtures

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-06.2a P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-18	BR-03 FR-01 to FR-04; approval fixtures
FR-06.3 P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-14, UC-15, UC-18, UC-20	BR-15 FR-01 to FR-04; approval fixtures
FR-06.3a P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-14, UC-15, UC-18, UC-20	BR-15 FR-01 to FR-04; approval fixtures
FR-06.4 P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-15, UC-16	BR-05, BR-15 FR-01 to FR-04; approval fixtures
FR-06.4a P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-15, UC-16	BR-05, BR-15 FR-01 to FR-04; approval fixtures
FR-06.4b P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-15, UC-16	BR-05, BR-15 FR-01 to FR-04; approval fixtures
FR-06.5 P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-13, UC-16, UC-19	BR-05 FR-01 to FR-04; approval fixtures
FR-06.5a P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-13, UC-16, UC-19	BR-05 FR-01 to FR-04; approval fixtures
FR-06.5b P0	S1, S2, S3, S9-S11; INT-02, INT-03	UC-13, UC-16, UC-19	BR-05 FR-01 to FR-04; approval fixtures
FR-07.1 P0	S1, S2, S3; INT-02, INT-04	UC-17, UC-18	BR-08 FR-01, FR-02, FR-04, FR-06
FR-07.1a P0	S1, S2, S3; INT-02, INT-04	UC-17, UC-18	BR-08 FR-01, FR-02, FR-04, FR-06
FR-07.1b P0	S1, S2, S3; INT-02, INT-04	UC-17, UC-18	BR-08 FR-01, FR-02, FR-04, FR-06
FR-07.1c P0	S1, S2, S3; INT-02, INT-04	UC-17, UC-18	BR-08 FR-01, FR-02, FR-04, FR-06
FR-07.2 P0	S1, S2, S3; INT-02, INT-04	UC-17, UC-18	BR-08, BR-15 FR-01, FR-02, FR-04, FR-06
FR-07.2a P0	S1, S2, S3; INT-02, INT-04	UC-17, UC-18	BR-08, BR-15 FR-01, FR-02, FR-04, FR-06
FR-07.2b P0	S1, S2, S3; INT-02, INT-04	UC-17, UC-18	BR-08, BR-15 FR-01, FR-02, FR-04, FR-06
FR-07.3 P0	S1, S2, S3; INT-02, INT-04	UC-19, UC-20	BR-03, BR-07 FR-01, FR-02, FR-04, FR-06
FR-07.3a P0	S1, S2, S3; INT-02, INT-04	UC-19, UC-20	BR-03, BR-07 FR-01, FR-02, FR-04, FR-06
FR-07.3b P0	S1, S2, S3; INT-02, INT-04	UC-19, UC-20	BR-03, BR-07 FR-01, FR-02, FR-04, FR-06

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-07.4 P0	S1, S2, S3; INT-02, INT-04	UC-21	BR-04, BR-07 FR-01, FR-02, FR-04, FR-06
FR-07.4a P0	S1, S2, S3; INT-02, INT-04	UC-21	BR-04, BR-07 FR-01, FR-02, FR-04, FR-06
FR-07.4b P0	S1, S2, S3; INT-02, INT-04	UC-21	BR-04, BR-07 FR-01, FR-02, FR-04, FR-06
FR-07.5 P0	S1, S2, S3; INT-02, INT-04	UC-14, UC-22	BR-07 FR-01, FR-02, FR-04, FR-06
FR-07.6 P0	S1, S2, S3; INT-02, INT-04	UC-22, UC-23	BR-07, BR-09 FR-01, FR-02, FR-04, FR-06
FR-07.6a P0	S1, S2, S3; INT-02, INT-04	UC-22, UC-23	BR-07, BR-09 FR-01, FR-02, FR-04, FR-06
FR-07.6b P0	S1, S2, S3; INT-02, INT-04	UC-22, UC-23	BR-07, BR-09 FR-01, FR-02, FR-04, FR-06
FR-08.1 P0	S1, S2, S3; INT-04	UC-06, UC-23, UC-25, UC-29, UC-41	BR-02 FR-02, FR-06, FR-07, FR-09
FR-08.2 P0	S1, S2, S3; INT-04	UC-27	BR-09 FR-02, FR-06, FR-07, FR-09
FR-08.3 P0	S1, S2, S3; INT-04	UC-28	BR-04, BR-09 FR-02, FR-06, FR-07, FR-09
FR-08.4 P0	S1, S2, S3; INT-04	UC-29	BR-02, BR-03, BR-09 FR-02, FR-06, FR-07, FR-09
FR-08.4a P0	S1, S2, S3; INT-04	UC-29	BR-02, BR-03, BR-09 FR-02, FR-06, FR-07, FR-09
FR-08.4b P0	S1, S2, S3; INT-04	UC-29	BR-02, BR-03, BR-09 FR-02, FR-06, FR-07, FR-09
FR-09.1 P0	S1, S2, S3; INT-04	UC-24	BR-04 FR-01, FR-02; condition fixtures
FR-09.1a P0	S1, S2, S3; INT-04	UC-24	BR-04 FR-01, FR-02; condition fixtures
FR-09.1b P0	S1, S2, S3; INT-04	UC-24	BR-04 FR-01, FR-02; condition fixtures
FR-09.1c P0	S1, S2, S3; INT-04	UC-24	BR-04 FR-01, FR-02; condition fixtures
FR-09.2 P0	S1, S2, S3; INT-04	UC-25	BR-01, BR-02 FR-01, FR-02; condition fixtures
FR-09.2a P0	S1, S2, S3; INT-04	UC-25	BR-01, BR-02 FR-01, FR-02; condition fixtures

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-09.3 P0	S1, S2, S3; INT-04	UC-26	BR-04 FR-01, FR-02; condition fixtures
FR-09.3a P0	S1, S2, S3; INT-04	UC-26	BR-04 FR-01, FR-02; condition fixtures
FR-09.4 P0	S1, S2, S3; INT-04	UC-25, UC-26, UC-27	BR-09, BR-10 FR-01, FR-02; condition fixtures
FR-10.1 P1	S1, S2, S3; INT-05	UC-30	BR-10, BR-12 FR-01, FR-02, FR-12
FR-10.2 P1	S1, S2, S3; INT-05	UC-30	BR-10 FR-01, FR-02, FR-12
FR-10.2a P1	S1, S2, S3; INT-05	UC-30	BR-10 FR-01, FR-02, FR-12
FR-10.3 P1	S1, S2, S3; INT-05	UC-30	BR-11 FR-01, FR-02, FR-12
FR-10.3a P1	S1, S2, S3; INT-05	UC-30	BR-11 FR-01, FR-02, FR-12
FR-11.1 P1	S1, S2, S3; INT-01, INT-05	UC-31	BR-01, BR-04, BR-10 FR-03 to FR-05, FR-12
FR-11.2 P1	S1, S2, S3; INT-01, INT-05	UC-31	BR-10 FR-03 to FR-05, FR-12
FR-11.3 P1	S1, S2, S3; INT-01, INT-05	UC-31	BR-10, BR-12 FR-03 to FR-05, FR-12
FR-11.4 P1	S1, S2, S3; INT-01, INT-05	UC-31	BR-10 FR-03 to FR-05, FR-12
FR-11.4a P1	S1, S2, S3; INT-01, INT-05	UC-31	BR-10 FR-03 to FR-05, FR-12
FR-11.4b P1	S1, S2, S3; INT-01, INT-05	UC-31	BR-10 FR-03 to FR-05, FR-12
FR-12.1 P1	S1, S2, S3; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	BR-10 FR-01; AI metadata and output fixtures
FR-12.1a P1	S1, S2, S3; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	BR-10 FR-01; AI metadata and output fixtures
FR-12.1b P1	S1, S2, S3; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	BR-10 FR-01; AI metadata and output fixtures
FR-12.2 P1	S1, S2, S3; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	BR-10 FR-01; AI metadata and output fixtures
FR-12.2a P1	S1, S2, S3; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	BR-10 FR-01; AI metadata and output fixtures

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-12.3 P1	S1, S2, S3; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	BR-02, BR-12 FR-01; AI metadata and output fixtures
FR-12.4 P1	S1, S2, S3; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	BR-08, BR-10 FR-01; AI metadata and output fixtures
FR-12.4a P1	S1, S2, S3; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	BR-08, BR-10 FR-01; AI metadata and output fixtures
FR-13.1 P1	S1, S2, S3; INT-04, INT-05	UC-32	BR-09, BR-10 FR-02, FR-08, FR-09, FR-12
FR-13.2 P1	S1, S2, S3; INT-04, INT-05	UC-32	BR-09, BR-10 FR-02, FR-08, FR-09, FR-12
FR-13.2a P1	S1, S2, S3; INT-04, INT-05	UC-32	BR-09, BR-10 FR-02, FR-08, FR-09, FR-12
FR-14.1 P1	S1, S2, S3; INT-08	UC-33	BR-01, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.1a P1	S1, S2, S3; INT-08	UC-33	BR-01, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.1b P1	S1, S2, S3; INT-08	UC-33	BR-01, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.1c P1	S1, S2, S3; INT-08	UC-33	BR-01, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.1d P1	S1, S2, S3; INT-08	UC-33	BR-01, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.1e P1	S1, S2, S3; INT-08	UC-33	BR-01, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.1f P1	S1, S2, S3; INT-08	UC-33	BR-01, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.2 P1	S1, S2, S3; INT-08	UC-33	BR-06, BR-10 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.2a P1	S1, S2, S3; INT-08	UC-33	BR-06, BR-10 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.2b P1	S1, S2, S3; INT-08	UC-33	BR-06, BR-10 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.3 P1	S1, S2, S3; INT-08	UC-33	BR-11, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-14.3a P1	S1, S2, S3; INT-08	UC-33	BR-11, BR-12 FR-01, FR-04, FR-05, FR-08, FR-12
FR-15.1 P1	S1, S2, S3; INT-09	UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-15.1a P1	S1, S2, S3; INT-09	UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.1b P1	S1, S2, S3; INT-09	UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.1c P1	S1, S2, S3; INT-09	UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.1d P1	S1, S2, S3; INT-09	UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.1e P1	S1, S2, S3; INT-09	UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.1f P1	S1, S2, S3; INT-09	UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.1g P1	S1, S2, S3; INT-09; S16	UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence; desktop carbon fixture in 3.4.3
FR-15.2 P1	S1, S2, S3; INT-09	UC-34	BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.2a P1	S1, S2, S3; INT-09	UC-34	BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.3 P1	S1, S2, S3; INT-09	UC-34, UC-35, UC-36	BR-02, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.3a P1	S1, S2, S3; INT-09	UC-34, UC-35, UC-36	BR-02, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.3b P1	S1, S2, S3; INT-09	UC-34, UC-35, UC-36	BR-02, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.4 P1	S1, S2, S3; INT-09	UC-34, UC-35, UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.4a P1	S1, S2, S3; INT-09	UC-34, UC-35, UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.4b P1	S1, S2, S3; INT-09	UC-34, UC-35, UC-36	BR-12, BR-13 FR-07 to FR-09; section 3.4 evidence
FR-15.5 P1	S1, S2, S3; INT-09	UC-34, UC-35	BR-01, BR-02 FR-07 to FR-09; section 3.4 evidence
FR-15.5a P1	S1, S2, S3; INT-09	UC-34, UC-35	BR-01, BR-02 FR-07 to FR-09; section 3.4 evidence
FR-15.5b P1	S1, S2, S3; INT-09	UC-34, UC-35	BR-01, BR-02 FR-07 to FR-09; section 3.4 evidence

Requirement / priority	Source / finding	Use cases	Rules and verification prerequisites
FR-16.1 P0	S1, S2, S3; INT-10	UC-15, UC-16, UC-17, UC-21, UC-24, UC-39	BR-01, BR-14 FR-01; section 2.8 event fixtures
FR-16.2 P0	S1, S2, S3; INT-10	UC-39	BR-14 FR-01; section 2.8 event fixtures
FR-16.2a P0	S1, S2, S3; INT-10	UC-39	BR-14 FR-01; section 2.8 event fixtures
FR-16.2b P0	S1, S2, S3; INT-10	UC-39	BR-14 FR-01; section 2.8 event fixtures
FR-16.2c P0	S1, S2, S3; INT-10	UC-39	BR-14 FR-01; section 2.8 event fixtures
FR-16.3 P0	S1, S2, S3; INT-10	UC-03, UC-04, UC-05, UC-07, UC-12, UC-40	BR-02, BR-15 FR-01; section 2.8 event fixtures
FR-16.4 P0	S1, S2, S3; INT-10	UC-40, UC-41	BR-01, BR-02 FR-01; section 2.8 event fixtures
FR-16.4a P0	S1, S2, S3; INT-10	UC-40, UC-41	BR-01, BR-02 FR-01; section 2.8 event fixtures
FR-17.1 P1	S1, S2, S3	UC-37	BR-01 FR-07 to FR-09, FR-12, FR-15
FR-17.2 P1	S1, S2, S3	UC-38	BR-10, BR-12 FR-07 to FR-09, FR-12, FR-15
FR-17.3 P1	S1, S2, S3	UC-37, UC-38	BR-11 FR-07 to FR-09, FR-12, FR-15
FR-17.3a P1	S1, S2, S3	UC-37, UC-38	BR-11 FR-07 to FR-09, FR-12, FR-15
FR-17.3b P1	S1, S2, S3	UC-37, UC-38	BR-11 FR-07 to FR-09, FR-12, FR-15

Acceptance-check IDs follow the AC- convention above. All 158 functional checks are planned; none is recorded as passed.

A.2 Nonfunctional requirements

Requirement / scope	Source / rationale	Linked use cases	Rules and verification prerequisites
NFR-01 Essential	S1; prototype target	UC-10	Search dataset; five-user workload
NFR-16 Essential	S1, S9; INT-04	UC-08, UC-18, UC-21, UC-24, UC-26, UC-28, UC-29	BR-04, BR-09, BR-10 BR-04, BR-09; clearance and unsafe fixtures
NFR-16a Essential	S1, S9; INT-04	UC-08, UC-18, UC-21, UC-24, UC-26, UC-28, UC-29	BR-04, BR-09, BR-10 BR-04, BR-09; clearance and unsafe fixtures

Requirement / scope	Source / rationale	Linked use cases	Rules and verification prerequisites
NFR-03 Essential	S1; BR-01	UC-01, UC-02, UC-04	Role matrix; hosted test environment
NFR-03a Essential	S1; BR-01	UC-01, UC-02, UC-04	Role matrix; hosted test environment
NFR-04 Essential	S1; security constraint	UC-01, UC-03	Framework credential setup; test secrets
NFR-04a Essential	S1; security constraint	UC-01, UC-03	Framework credential setup; test secrets
NFR-13 Essential	S1; S15; privacy constraint	Cross-cutting: personal-data fixtures, AI, exports and backups	Fictional manifest; outbound capture; role fixtures; retention clock; real-data gate
NFR-13a Essential	S1; S15; privacy constraint	Cross-cutting: personal-data fixtures, AI, exports and backups	Fictional manifest; outbound capture; role fixtures; retention clock; real-data gate
NFR-13b Essential	S1; S15; privacy constraint	Cross-cutting: personal-data fixtures, AI, exports and backups	Fictional manifest; outbound capture; role fixtures; retention clock; real-data gate
NFR-13c Essential	S1; S15; privacy constraint	Cross-cutting: personal-data fixtures, AI, exports and backups	Fictional manifest; outbound capture; role fixtures; retention clock; real-data gate
NFR-13d Essential	S1; S15; privacy constraint	Cross-cutting: personal-data fixtures, AI, exports and backups	Fictional manifest; outbound capture; role fixtures; retention clock; real-data gate
NFR-02 Essential	S1; INT-06	UC-10, UC-11, UC-14	Core task scripts; five test users
NFR-05 Essential	S1; INT-02	UC-13, UC-18, UC-19, UC-21, UC-22, UC-23	FR-06, FR-07; concurrency/failure fixtures
NFR-05a Essential	S1; INT-02	UC-13, UC-18, UC-19, UC-21, UC-22, UC-23	FR-06, FR-07; concurrency/failure fixtures
NFR-06 Essential	S1; INT-06	UC-30, UC-31, UC-32, UC-33, UC-38	AI delay/failure controls; manual paths
NFR-06a Essential	S1; INT-06	UC-30, UC-31, UC-32, UC-33, UC-38	AI delay/failure controls; manual paths
NFR-07 Essential	S1; evaluation target	UC-31	FR-11; ten held-out request labels
NFR-08 Essential	S1; evaluation target	UC-30	FR-10; fifty held-out asset labels
NFR-09 Essential	S1; INT-05	UC-30, UC-31, UC-32, UC-33, UC-38	FR-12, FR-17; twenty outputs

Requirement / scope	Source / rationale	Linked use cases	Rules and verification prerequisites
NFR-10 Essential	S1; recovery constraint	UC-03, UC-22, UC-29	Protected backup and restore environment
NFR-10a Essential	S1; recovery constraint	UC-03, UC-22, UC-29	Protected backup and restore environment
NFR-11 Essential	S1; INT-06; S14	Cross-cutting: all delivered pages and complete processes	WCAG 2.1 A/AA audit matrix; automated and manual evidence
NFR-12 Essential	S1; reproducibility constraint	UC-03, UC-37	README, sample data and clean environment
NFR-12a Essential	S1; reproducibility constraint	UC-03, UC-37	README, sample data and clean environment
NFR-14 Essential	S3; budget target	UC-33	Quota controls; allowance and spending ledger
NFR-14a Essential	S3; budget target	UC-33	Quota controls; allowance and spending ledger
NFR-14b Essential	S3; budget target	UC-33	Quota controls; allowance and spending ledger
NFR-15 Essential	S1; INT-09	UC-34, UC-35, UC-36, UC-37, UC-38	FR-15, FR-17; twenty transfer/service examples
NFR-15a Essential	S1; INT-09	UC-34, UC-35, UC-36, UC-37, UC-38	FR-15, FR-17; twenty transfer/service examples
NFR-17 Essential	S1; INT-07; prototype target	UC-31	BR-01, BR-10, BR-12 FR-11, FR-12; five matched request pairs

All 31 nonfunctional checks are planned. Numerical targets are prototype acceptance obligations, not reported results or institutional service guarantees.

Appendix B Use case actor index

This index supplements the complete overview in section 6 (Figure 1). It preserves all 41 user goals and copies the primary and supporting actors from section 7. Actor participation does not grant permissions or replace the approval sequence. Asset administrator is the shortened name for Asset-management administrator in section 2.5; External AI service is a supporting system, not a business decision-maker.

Use case	Primary actor	Supporting actors
UC-01 Sign in	Campus user	None
UC-02 Sign out	Campus user	None
UC-03 Manage user accounts	System administrator	None
UC-04 Assign roles and scope	System administrator	None
UC-05 Manage departments	System administrator	None
UC-06 Register an asset	Asset custodian	None
UC-07 Update an asset record	Asset custodian	None
UC-08 Publish available stock	Asset custodian	None
UC-09 Withdraw a listing	Asset custodian	None
UC-10 Search available assets	Campus user	None
UC-11 Record resource demand	Requester or representative	None
UC-12 Amend resource demand	Requester or representative	None
UC-13 Cancel remaining demand	Requester or representative	None
UC-14 Track a request	Requester or representative	None
UC-15 Submit an allocation request	Requester or representative	None
UC-16 Cancel a pending request	Requester or representative	None
UC-17 Review destination acceptance	Destination approver	None
UC-18 Approve release and reserve	Asset custodian	None
UC-19 Cancel an approved reservation	Requester or representative	Asset custodian
UC-20 Arrange a transfer	Asset custodian	Receiving custodian
UC-21 Dispatch a transfer	Asset custodian	None
UC-22 Confirm receipt	Receiving custodian	None
UC-23 Resolve a transfer discrepancy	Asset administrator	Asset custodian, Receiving custodian
UC-24 Report an asset defect	Campus user	None
UC-25 Record maintenance work	Maintenance staff	None

Use case	Primary actor	Supporting actors
UC-26 Complete inspection and clearance	Maintenance staff	None
UC-27 Propose an asset action	Asset custodian	Sustainability officer
UC-28 Review a disposition proposal	Asset administrator	None
UC-29 Record disposition execution	Asset custodian	None
UC-30 Review AI classification	Asset custodian	External AI service
UC-31 Review semantic matches	Requester or representative	External AI service
UC-32 Review sustainable actions	Sustainability officer	Asset custodian, External AI service
UC-33 Use the campus assistant	Campus user	External AI service
UC-34 Validate financial benefit	Finance officer	Procurement officer
UC-35 Validate environmental evidence	Sustainability officer	None
UC-36 View sustainability indicators	Report viewer	None
UC-37 Generate an operational report	Report viewer	None
UC-38 Review an AI report summary	Report viewer	External AI service
UC-39 Read notifications	Campus user	None
UC-40 Search audit records	Asset administrator	None
UC-41 Inspect asset history	Campus user	None

The detailed preconditions, alternatives, exceptions and outcomes in section 7 remain authoritative when interpreting an association.